Design and Application of a Performance Analysis Tool to Maximise Range, Endurance and Stability of a Jet-Powered Tailless Uncrewed Aerial System (UAS) for UK Civil Aviation Authority (CAA) Open Category Operations

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Nomenclature

Symbol	**Description**
AR	Aspect ratio, [–]
λ	Taper ratio ($c_{\text{tip}}/c_{\text{root}}$) [–]
$\alpha_{0,\text{root}}$, $\alpha_{0,\text{tip}}$	Zero-lift angles at root and tip [deg]
ϕ_G	Geometric washout (wing twist) [deg]
$\Lambda_{C/4}$	Quarter-chord sweep angle [deg]
C_D	Overall drag coefficient [–]
$C_{D,0}$	Zero-lift (parasite) drag coefficient [–]
$C_{D,\text{min}}$	Minimum drag coefficient [–]
C_L	Lift coefficient [–]
$C_{L,\text{max}}$	Maximum lift coefficient [–]
$C_{L,\text{TO}}$	Lift coefficient during take-off [–]
$C_{L,\text{LDG}}$	Lift coefficient during landing [–]
C_m	Pitching moment coefficient [–]
$C_{m,0}$	Zero-lift pitching moment coefficient [–]
c_{tp}	TSFC parameter used in MATLAB code [kg/s/N)]
D	Aerodynamic drag force [N]
E	Endurance (flight time) [s or min]
FF	Fuel flow [$\mathrm{N} \cdot \mathrm{h}^{1}$]
F_N	Net thrust force [N]
g	Gravitational acceleration, 9.81 $\mathrm{m/s^2}$
K	Lift-induced drag factor [–]
L	Aerodynamic lift force [N]
M	Mach number [–]
n	Load factor, $1/\cos\theta$ [–]
q	Dynamic pressure,
R	Maximum range [m]

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Symbol	**Description**
S	Wing reference area [m^2]
S_{TG}	Take-off ground roll distance [m]
S_{LDG}	Landing distance [m]
SAR	Specific air range, distance per unit fuel weight [m/kg]
SE	Specific endurance, time per unit fuel weight [min/kg]
T	Thrust [N]
T_R	Required thrust in level flight [N]
$\frac{T}{W}$	Thrust-to-weight ratio [–]
τ	Free-roll time before braking begins [s]
V	Flight velocity [m/s]
V_∞	Freestream (cruise) airspeed [m/s]
V_V	Vertical climb rate [m/s]
W	Aircraft weight [N]
W_0	Initial take-off weight [N]
W_i	Final in-flight weight [N]
W_{empty}	Empty weight [N]
W_f	Fuel weight burned [N]
$\frac{W}{S}$	Wing loading [N/m^2]
$\rho,\ \rho_\infty$	Air density [kg/m^3]
ρ_0	Sea-level air density (1.225 kg/m^3)

Abstract

This dissertation addresses the growing demand for efficient Uncrewed Aerial Systems (UAS) in civil and commercial applications, ranging from environmental monitoring to search and rescue missions. The primary aim of this study is the development and application of an integrated MATLAB, XFLR5 and FLOW5 toolchain for the preliminary design and performance assessment. The design configuration consists of a tailless jet-powered UAS constrained under UK CAA Open Category regulations, with the primary objective of maximising range, endurance and stability metrics.

The workflow begins with a constraints analysis to compute the aircraft's initial sizing and feasible design point regions. This is subsequently followed by an aerofoil comparability analysis resulting in a 3D flying wing configuration with the MH60 profile paired with thicker aerofoil sections serving as a fuselage. The stability analysis involved iteratively adjusting the wing's geometry. The final design achieved sufficient damping ratios for oscillatory modes (Phugoid: 0.005, Short-period: 0.323, Dutch-Roll: 0.05) and adequate time to halves for non-oscillatory modes (Roll Mode: 0.440, Spiral Mode: 13.933). The final stable design was then further analysed, yielding a lift-to-drag ratio of 23.5 at 6° angle of attack. Trimmed flight occurred at lift coefficient of 0.24 and 3.9° angle of attack.

The propulsion evaluation led to selecting the XICOY X45 micro turbine due its low thrust specific fuel consumption at cruise speeds that maximise endurance and range (28 and 38 m/s, respectively). At maximum take-off weight of 25 kg and 4 L fuel capacity, performance calculations resulted in a specific air range (SAR) of ∼30,000 m/kg (∼96 km total range) and specific endurance (SE) of ∼17 min/kg (∼54 min total airtime). The toolchain also computed calculate ground performance: take-distance of ∼158 m and a landing distance of ∼246 m at a touchdown velocity of ∼17 m/s. Bridging spreadsheet methods with high-fidelity CFD, this fast and low-cost framework accelerates the early-stage UAS design process. Further establishing a foundation for future enhancements ranging from structural optimisation and CFD validations to adaptations for electric or conventional fixed-wing configurations.

Lay Summary

This project aims to design a small jet-powered drone, with the purpose of having enhanced flight durations and distance covered, while complying with the UK's aviation safety regulations. The drone configuration features a single wing as its main body, commonly called a 'flying wing'. The design also lacks the tail fins seen on most aircraft, thereby making it lighter and more efficient. Computer simulations are used to test numerous wing configurations and engines to find the best trade-off for stable, long-distance flight. The final design must weigh less than 25 kg and fly below 120 m above sea level, with the capacity to carry sensors for a number of applications. While the followed regulations for this design mean the drone must stay within the operator's line of sight, the design and performance tool could still be useful in the future for missions like environmental monitoring or disaster response, where flying for longer periods would allow more data to be gathered. A future step would be to build and test a physical prototype to confirm its performance.

Declaration of Originality

I hereby declare that the work presented in this report is all my own work. All work that has been taken from other papers or books has been correctly referenced. If any work is found that has not been referenced, contact me immediately, and it will be amended.

Copyright Statement

Acknowledgments

I would like to express my sincere gratitude to my supervisor, Mr Khristopher Kabbabe, for their continuous guidance, support, and constructive feedback throughout the course of this project. I would also like to thank Kushal Singh, Teaching Assistant, for their helpful input and assistance during the design and analysis phases of the work.

1 Introduction

Uncrewed Aerial Systems (UAS) have undergone extensive development in the last decade or so, with applications ranging from environmental monitoring to precision agriculture, infrastructure assessment, and search-and-rescue missions, to name a few (NPSA, 2023). For the most part, the industry has followed conventional designs, with a distinct fuselage and tailplane. However, advancements in computational design and light propulsion technologies have enabled exploring unconventional designs, of which the flying-wing configuration is an example. Incorporating payload space within the wing structure and removing individual tail surfaces enables flying wings to produce significantly higher lift-to-drag ratios and reduced structural weight. Consequently, this enables flying wings to achieve improved range and endurance compared to conventional aircraft configurations (Gudmundsson, 2013).

Concurrently, small jet propulsion is becoming a credible alternative to electric and piston-based propulsion systems. Modern miniature turbojets are demonstrating thrust-to-weight ratios exceeding those of piston or electric engines of similar size. This enables higher cruise speeds and longer loiter times for missions demanding quick deployment, coverage or extended flight durations (Yi, 2019). However, integrating a jet engine into a UAS with a take-off weight of less than 25 kg, as required by regulations introduced by the UK Civil Aviation Authority (CAA) within the Open Category, presents several challenges. The CAA's Open Category regulations restrict operations to Visual-Line-of-Sight (VLOS) and below a height of 120 m (CAA, 2025). Since this design is intended for the Open Category, it makes it inherently more challenging and potentially less suited for real world applications. In addition, compliance with these regulations requires careful optimisation of the propulsion system, aerodynamic efficiency, and flight stability. The primary purpose of this project is to test the performance tools and validate the design process.

Despite extensive foundational studies regarding flying-wing aerodynamics (such as Milligan, 2000) and stand-alone studies of small-jet propulsion, there is an evident scarcity of literature in jet-powered, tailless flying-wing UAS designs at small scales. Existing literature tends to focus on electrically powered model aircraft or heavier, military-class flying-wing configurations above Civil Aviation Authority (CAA) guidelines. Moreover, there is little adaptation of standard performance analysis tools to account for the complex interactions between tailless structures, aerofoil profiles, and propulsion system trade-offs. Consequently, initial-stage designers lack a tailored performance analysis framework to investigate the overall design range under regulatory constraints, which is another driving factor motivating this research.

By integrating aerodynamics, propulsion, and regulation aspects as a complete toolchain, the research offers an operational process to contribute to unconventional UAS design literature. The research offers valuable insights to small-scale UAS application and development organisations that wish to study tailless aircraft configurations as well as hobbyists. The toolchain further bridges conceptual studies into real-world applications that could link to environmental monitoring organisations, disaster-response teams and businesses focused on missions requiring enhanced endurance and range.

Aims and Objectives

Aims: Development of a performance analysis tool to maximise range and endurance of a jet-powered UAS. Further applying the tool to a case study focusing on the design and optimisation of a tailless flying-wing configuration, ensuring compliance with UK Civil Aviation Authority (CAA) Open Category constraints.

Objectives:

1. **Design Requirement Specification:** Evaluate the UK CAA Open Category constraints (Maximum Take-off Weight (MTOW) $\leq 25\,\text{kg}$, service ceiling $120\,\text{m}$ and translate these into quantitative design requirements for wing loading and thrust-to-weight ratio.

2. **Aerodynamic Configuration Optimisation:** Analyse multiple aerofoil sections using XFLR5 to identify profiles that maximise lift-to-drag ratio and inherent stability and further perform 3D wing modelling and analysis in FLOW5.

3. **Geometry and Stability Optimisation:** Iteratively adjust wing area, aspect ratio, sweep, taper, and centre-of-gravity location to satisfy stability margins within the $25\,\text{kg}$ weight constraint. Calculate stability derivatives (e.g. C_{m_α}, C_{n_β}) and dynamic mode characteristics (short-period, phugoid, Dutch roll) to verify the aircraft is both statically and dynamically stable.

4. **Propulsion System Choice:** Evaluate different compact sized jet engines, calculate thrust-specific fuel consumption (TSFC) curves, and select the engine offering the lowest TSFC at the cruise thrust level.

5. **Performance Modelling:** Develop a MATLAB-based tool that combines aerodynamic parameters and propulsion data to compute range, endurance, take-off/landing distances, and stability margins using fundamental performance equations (e.g. Breguet Range).

6. **Final Design Validation:** Demonstrate that the optimised UAS configuration meets all aim-related performance metrics and regulatory requirements, and evaluate final design operating ranges.

2 Literature Review

2.1 Introduction

Uncrewed Aerial Systems (UAS) have evolved quickly, and contemporary designs are venturing into non-traditional configurations. Tailless flying wings are becoming increasingly popular due to their high aerodynamic efficiency and minimal drag. Traditionally reserved for niche applications such as the Northrop B-2 Spirit, more recent research by NASA and MIT has pointed to new forms of the flying wing that are lighter and enable more energy-efficient flights (L. Chandler, 2019).

Jet propulsion in UAS is also emerging; unlike electric or piston-powered UAVs, small turbojet engines can provide very high thrust-to-weight ratios. However, at sub-25 kg scale (a requirement by the UK CAA Open Category), jet engines can be challenging since miniature turbines were essentially unavailable until very recently (Adams, 2019). Moreover, while flying wings offer improved range and endurance through high aerodynamic efficiency, incorporating small jet engines require careful trade-off between performance, propulsion efficiency, and stability.

Although general design texts, such as Gudmundsson (2022) and Leishman (2019), establish the fundamentals of range, endurance, thrust-specific fuel consumption (TSFC), and flight mechanics, there remains a gap in applied research specifically tailored to jet-powered, tailless flying-wing UAS at small scales. Current literature lacks dedicated performance analysis tools that integrate both the aerodynamic complexities of tailless configurations and the specific operational challenges of jet propulsion at small scales. This literature review addresses this gap by examining existing research on tailless flying-wing configurations, small-scale jet propulsion, and performance analysis methodologies, with a focus on identifying the specific challenges and opportunities relevant to jet-powered UAS operating within CAA Open Category constraints.

2.2 Regulatory Context

In the United Kingdom, drone flights are regulated by the Civil Aviation Authority (CAA) into three categories: Open, Specific, and Certified. The Open Category is low-risk drone flight and covers most leisure and some commercial use. Pilots must meet some requirements and limitations to fly in this category for safety and compliance. These regulations detail a Maximum Take-Off Weight (MTOW) below 25 kilograms, inclusive of any payload and attached sensors (CAA, 2025). It is also required that flights are below a ceiling altitude of 120 meters (400 feet) mainly in place to minimise the possibility of collision with manned aircraft. Furthermore, operators are required to keep Visual Line of Sight (VLOS) with their drone throughout, thereby being in a position to keep an eye on its location and react swiftly to any possible dangers (CAA, 2025).

The Specific Category is used for drone flights that are not in the Open Category and require an operational authorisation from the CAA. The operators must explain the operations they wish to conduct and provide evidence they can do so lawfully and safely. Two types of authorisations that can be given are 'PDRA01' operational authorisation and 'Operating Safety Case' (OSC) authorisation (CAA, 2025). Approved applications would then allow flights beyond visual line of sight (BVLOS), necessary for real-world UAS applications such as environmental monitoring and search and rescue missions.

This project primarily focuses on a design within the Open Category regulation; however, if the specific design specifications exceed the limitations of this category, such as surpassing the 25 kg MTOW, further consideration of the Specific Category—particularly the OSC-based authorisation will be required to ensure compliance with regulatory standards.

2.3 Existing Research on Tailless Flying-Wings

Flying-wing designs trade conventional tail surfaces for extended wing area, which can enhance range and endurance. Comparative studies have found that blended-wing-body and flying-wing UAVs achieve higher lift-to-drag ratios and longer endurance than equivalent conventional designs, primarily due to their low parasitic drag and increased wing area. The flying wing configuration offers improved range and endurance performance due to these aerodynamic advantages, with many modern flying-wings, such as the Boeing X-48, capitalising on this benefit (Seyhun Durmuş, 2023). In design practice, tools like XFLR5 (based on XFoil) are frequently used to estimate flying-wing aerodynamics. Publications on flying-wing development, often electrically powered, report drag polars, lift coefficients, and optimal wing loading through combined CFD and flight testing (Chung, Ma, and Shiau, 2019).

The absence of a tail makes both static and dynamic stability a challenging task. Flying wings must achieve a stable state by other means, typically by employing swept wings, reflexed aerofoils, geometric twist, and centre of mass placement. To attain stability, an experimental low speed flying wing UAV used a 30° sweep and 2° dihedral. While dihedral and winglets increase lateral stability, sweep moves the aerodynamic centre further aft, aiding pitch stability (Chung, Ma and Shiau, 2019). Designers often select reflex-camber aerofoil profiles, exhibiting a nose-up pitching moment to help trim the aircraft. Win dihedral angle has a significant influence on lateral-directional stability, while other researchers tested reflex aerofoils and winglets to restore yaw and roll damping (Lei, 2014).

Overall, the tailless UAS literature indicates that flying wings can provide improved aerodynamic efficiency (range/endurance), but at the cost of complex stability considerations (Chung, Ma and Shiau, 2019). Most articles address specific topics (e.g. aerofoil optimisation, dihedral effects, and computational tool based trim analysis), but none provide overall analysis of performance and stability for small jet powered systems. Classic design sources such as Gudmundsson, 2022, and Leishman, 2022, form the foundations, but tailored methods for tailless cases remain underdeveloped.

2.4 Research Gaps in Tailless Flying Wing Performance Analysis

While there is a handful of research exists separately on flying-wing aerodynamics, stability and jet propulsion separately, the specific combination of jet-powered tailless flying wing design remains sparsely covered in literature. Most published research on flying-wing aircraft has employed electric propulsion systems or gliders, rather than examining the application of jet engine propulsion integration (Chung, Ma and Shiau, 2019). As a result, investigations into jet-powered UAVs tend to be focused on larger platforms exceeding the 25 kg CAA constraints or conceptual systems for the purpose of increasing battery endurance (Chandler, 2019). Detailed performance and stability investigations of tailless flying-wing aircraft are limited, with most investigations being on electrically powered prototypes and lacking data for small-scale, jet-powered configurations. Consequently, there is a clear gap in applied research pertaining to the performance, stability, and design optimisation of jet-powered, tailless UAS capable of flight within the CAA Open Category framework.

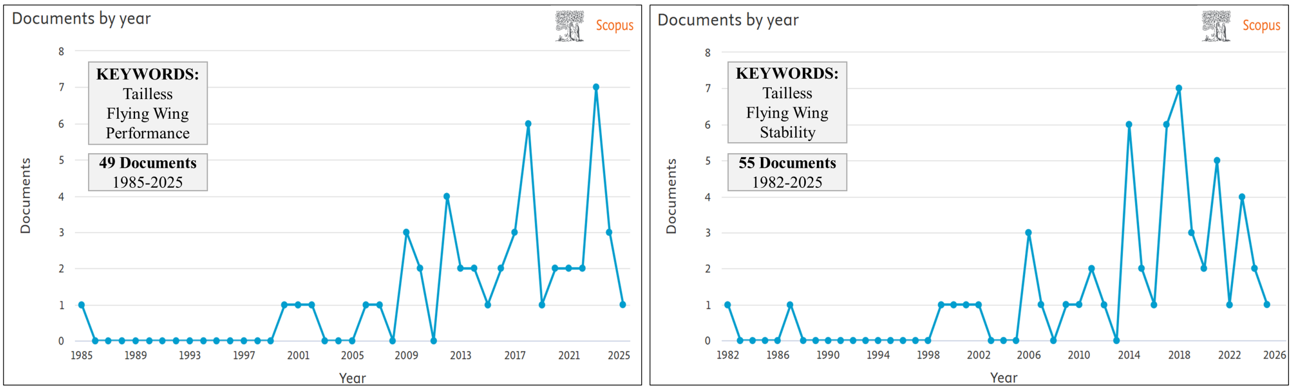


Fig. 2.1. Article and Reports by Year Linking to Tailless Flying Wings (Scopus, 2025).

The analysis of research documents per year (Figure 2.3) identifies 104 publications (Scopus, 2025) combining ”flying wing,” ”tailless,” and related keywords. This highlights a clear gap in studies addressing performance, stability, and design optimisation for jet-powered tailless flying wing, particularly within the CAA Open Category framework.

Furthermore, performance analysis tools specifically tailored for such configurations are limited. Existing performance tools lack analyses of characteristics such as optimal velocities for range and endurance, take-off and landing performance, stability configurations, and jet propulsion efficiency, particularly for tailless flying wings. This report aims to contribute to this area by providing insights into performance analysis tools and methodologies, facilitating further development in design optimisation and operational analysis for these configurations.

2.5 Theoretical Framework

2.5.1 Aerodynamics of a Flying Wing

The primary aerodynamic advantage of a tailless flying-wing design, especially for maximised endurance and range based missions, is its lower drag coefficient. By removing fuselage and tail, a flying wing can cut zero-lift-drag (C_{D0}) by up to 40% at the same aspect ratio. The drag savings directly affect the overall lift-to-drag ratio, offering more efficient cruise performance and improved climb capability. Furthermore, the elimination of tail surfaces results in a saving in structural weight, thereby lowering wing loading. Decrease in wing loading not only lowers bending moments, allowing a lighter wing structure, but also decreases the sink rate, which is one of the key parameters for long endurance missions (Milligan, 2000).

2.5.2 Constraints Analysis

Constraint analysis is a foundational method in aircraft conceptual design, used during initial sizing to determine wing loading (W/S) and thrust-to-weight ratio (T/W) that satisfy all performance requirements. By translating constraints such as service ceiling, take-off distance, and climb rate into mathematical relationships (T/W = f(W/S)), designers are able to plot these as isopleths on a constraint diagram. This graph maps W/S (x-axis) against T/W (y-axis), with each curve representing a performance limit. The feasible design space is located where all the constraints are satisfied simultaneously (Gudmundsson, 2022).

2.5.3 Reflexed Aerofoils and Wing Geometric Twist

Tailless designs require aerofoil shapes that produce a nose-up moment for trim. Reflexed (reverse-camber) aerofoils are therefore considered as their reflexed trailing edge generates a positive pitching moment that can replace a conventional tail (Leishman, 2025). Traditional cambered profiles, such as the NACA 4415, often have a large nose-down moment C_m $\approx$-0.10) at cruise lift coefficients. Reflexed aerofoil profiles such as the Horten Standard 12% thick 4% camber or MH60 10% are specially designed to have a near-zero pitching moment at similar conditions, thereby minimising trim drag and control effort (Gudmundsson, 2022).

Wing twist or geometric washout is a significant design parameter in tailless flying-wing design, its main purpose is to prevent tip stall, improve lateral and longitudinal stability, and maintain positive elevon control at high AoA. Washout affects both span-wise lift distribution and elevon alignment at cruise. As a means of early-stage design—particularly for radio-controlled (RC) flying wings—the Panknin formula represents a helpful method for estimating the washout required to ensure outboard elevons are aligned with the local flow. This semi-empirical method is a great starting point prior to detailed aerodynamic optimisation. The twist of the wing required to achieve trimmed flight is calculated by the Panknin Twist Formula as follows:

$$\phi_G = 71429 \cdot \frac{(K_1 C_{m,\text{root}} + K_2 C_{m,\text{tip}}) - C_{L,\text{cruise}} K_{\text{SM}}}{AR^{1.43} \Lambda_{C/4}} - (\alpha_{0,\text{root}} - \alpha_{0,\text{tip}}) \tag{1}$$

where

$$K_1 = \frac{3 + 2\lambda + \lambda^2}{4(1 + \lambda + \lambda^2)}, \quad K_2 = 1 - K_1, \quad \lambda = \frac{c_{tip}}{c_{root}}.$$

This allows a quantification of the geometric washout ϕ_G needed to balance aerodynamic moment contributions from root and tip aerofoils at a target cruise lift coefficient. The combination of reflex aerofoil and washout, adjusts the wing's lift and moment coefficients so that the aircraft can be trimmed without a tail while maximising aerodynamic efficiency (Gudmundsson, 2022).

2.5.4 Stability and Control in Tailless Flying-Wing Configurations

For an aircraft to maintain stable and controlled flight, it must return to its equilibrium condition after being subjected to a disturbance, across all three axes: pitch, roll, and yaw. If a gust or external force disturbs the aircraft and it fails to restore itself to its trimmed state, it leads to the operator no longer being in control of the aircraft (Gabe, 2024). A stable aircraft must converge back to its equilibrium with minimal oscillations, ensuring smooth and predictable behaviour.

In conventional aircraft, longitudinal stability is mainly provided by the horizontal stabiliser, which generates a restoring nose-down moment following pitch disturbances. However, achieving stability becomes particularly challenging in tailless flying-wing configurations. In a tailless flying wing, stability must be integrated into the aerodynamic design of the wing through techniques such as the use of reflexed aerofoils, geometric washout, and centre of mass location relative to the aerodynamic centre (Inchbald, 2025). These features collectively ensure that the aircraft produces a positive zero-lift pitching moment ($C_{m_0} > 0$). These wing design considerations ensure the aircraft can achieve the necessary negative longitudinal stability condition such that:

$$\frac{dC_m}{d\alpha} < 0,$$

ensuring that an increase in angle of attack results in a nose-down restoring moment allows the wing to trim itself at the required lift without the need for a separate horizontal stabiliser. Figure 2.2 illustrates the static stability conditions by showing the moment coefficient plotted against angle of attack (AOA). A negative slope indicates static stability, a positive slope indicates instability, and a zero gradient marks the neutral point. The trim point is located where $C_m = 0$ (Andreas, 2018).

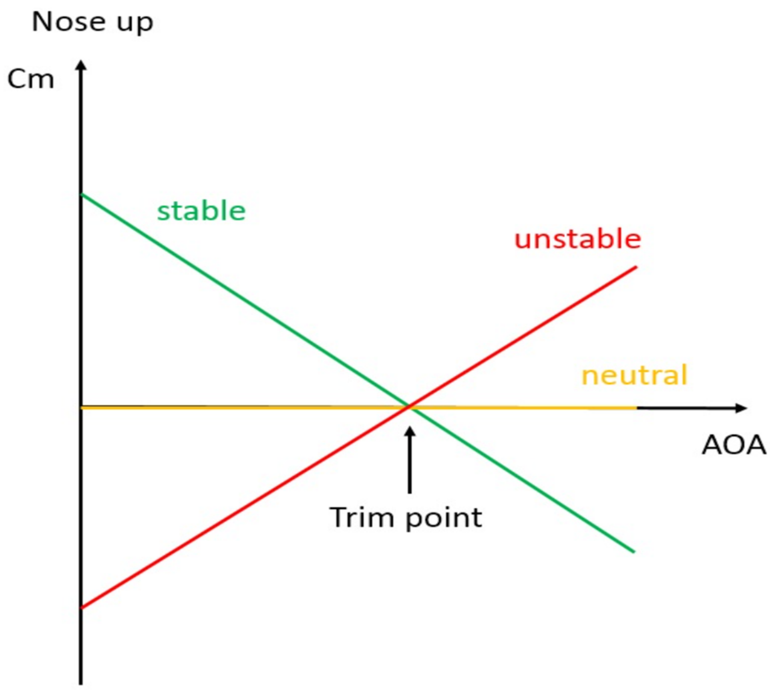


Fig. 2.2. Aircraft Stability Conditions: Pitching Moment Coefficient vs. AOA (Andreas, 2018).

Static stability describes the initial tendency of the aircraft to return to its trimmed condition after a disturbance. Positive static stability means the aircraft begins to return to equilibrium; negative static stability means the disturbance grows over time. For instance, in pitch, static stability requires that an increase in angle of attack produces a nose-down moment, driving the aircraft back toward its initial flight condition (Anon, 2004). However, static stability alone does not guarantee a desirable flight response. Dynamic stability describes the motion over time following a disturbance. Even a statically stable aircraft can exhibit dynamically unstable behaviour, such as oscillations that grow rather than decay (Leishman, 2025).

Figure 2.3 illustrates that an aircraft may be statically stable while still being dynamically neutral or unstable. A tailless flying wing, lacking the stabilising influence of a tailplane, must generate all its dynamic stability directly through the wing itself (Milligan, 2000). In practice, this means carefully managing each of the principal dynamic modes.

The typical longitudinal response of most aircraft to perturbations consists of two under damped oscillatory modes with distinctly different time scales. One is the short-period mode, which has a relatively brief period and is usually strongly damped, directly affecting handling qualities. The other is the phugoid mode, characterised by a much longer period and relatively light damping (Caughey, 2011).

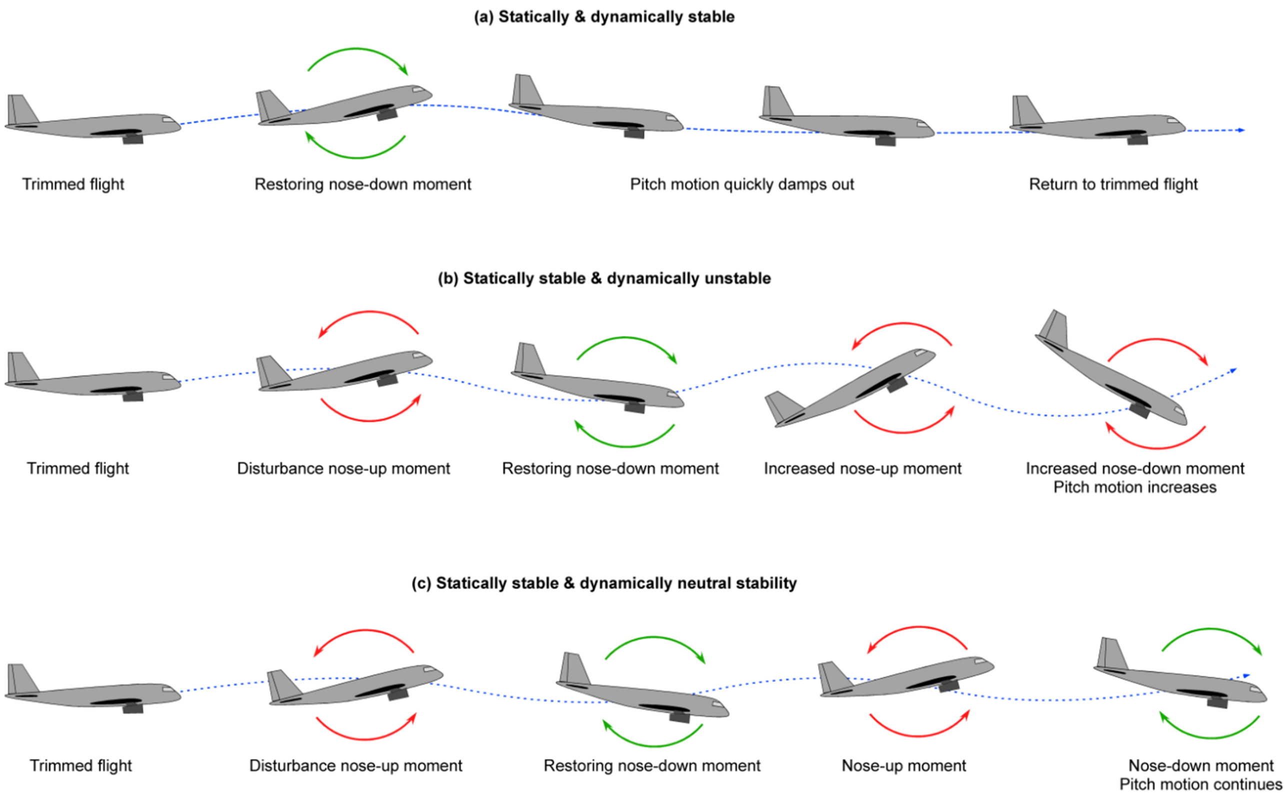


Fig. 2.3. Dynamic Modes (Leishman, 2025).

The typical lateral response of most aircraft to disruptions typically includes one damped oscillatory mode and two exponential modes. The oscillatory component, known as the Dutch roll mode, has a relatively short period and can be lightly damped, especially in swept-wing aircraft, producing a coupled motion of roll, sideslip, and yaw (Caughey, 2011). Among the exponential modes, the rolling mode is very heavily damped and primarily reflects the aircraft's response in roll. The second exponential mode, the spiral mode, may be stable or unstable but generally exhibits a long time constant, making it manageable for piloted aircraft even when marginally unstable (Caughey, 2011). Ensuring adequate damping and time constants in each of these modes is as crucial as static stability as they dictate whether the pilot retains control and the aircraft remains safe.

Furthermore, lateral stability in flying wings is typically improved using wing dihedral and sweep. Directional stability, which is inherently more difficult without a vertical fin, can be achieved using features such as winglets (Gurukul, 2018). Tailoring the wing geometry and CoM location is essential to maintain the balance between stability, control authority, and overall efficiency in flying wing designs.

Equally important are the static-stability design parameters and their "rule-of-thumb" targets. Longitudinal stability requires a positive static margin—i.e. the centre of gravity positioned ahead of the aerodynamic neutral point, such that

$$\frac{dC_m}{d\alpha} < 0$$

and any nose-up disturbance generates a restoring nose-down moment. Lateral stability is secured by ensuring

$$C_{L_\beta} < 0,$$

so that side-slips produce a restoring roll, while directional stability demands

$$C_{N_\beta} > 0,$$

so that side-slips also induce a restoring yawing moment. In a tailless configuration these derivatives are tuned through wing sweep, dihedral, geometric twist and, where necessary, small vertical surfaces or drag rudders at the winglets (Gabe, 2024). By balancing static margins, control authority, and modal damping, a tailless flying wing can achieve both the stability and manoeuvrability required for a safe flight.

2.5.5 Range and Endurance

Two parameters that are crucial for aircraft performance are the **range** and the **endurance**. The maximum range (R) is the greatest distance an aircraft can fly on a given fuel load. This is derived from the Breguet Range Equation for jet-powered aircraft, as shown in Equation ((2)).

$$R_{\text{Jet}} = \frac{V_\infty}{c_{\text{th}}} \cdot \frac{L}{D} \cdot \ln\left(\frac{W_0}{W_i}\right) \tag{2}$$

This equation suggests that maximum range is achieved by maximizing both the velocity V_∞ and the lift-to-drag ratio (L/D). However, since L/D is a function of angle of attack, which itself varies with V_∞ in level flight, these variables are not independent. To maximise range, the flight condition should satisfy a maximum value of $V_\infty \cdot (L/D)$ (Anderson, 1999).

For optimal range performance in jet aircraft:

- The engine should operate at or near the minimum thrust-specific fuel consumption (TSFC)
- Flight occurs at high altitudes where jet engines are most efficient
- The aircraft carries the maximum feasible fuel volume
- Flight is conducted at or near the aircraft's maximum $C_L^{1/2}/C_D$

To further examine the product of V_∞ and L/D in steady, level flight:

$$L = W = \frac{1}{2}\rho_\infty V_\infty^2 S C_L \tag{3}$$
$$\Rightarrow V_\infty = \sqrt{\frac{2W}{\rho_\infty S C_L}}$$

Substituting into the expression for $V_\infty \cdot (L/D)$ gives:

$$V_\infty \cdot \frac{L}{D} = \sqrt{\frac{2W}{\rho_\infty S}} \cdot \frac{C_L^{1/2}}{C_D} \tag{4}$$

According to this modified Breguet Range equation for jet powered systems, the maximum range condition is:

$$R_{\text{Jet}} = \frac{2}{c_{\text{th}}}\sqrt{\frac{2}{\rho_\infty S}}\left(\frac{C_L^{1/2}}{C_D}\right)_{\text{max}}\left(\sqrt{W_0} - \sqrt{W_{\text{empty}}}\right) \tag{5}$$

Theoretical maximum range is achieved when the aircraft flies at a velocity where zero-lift drag equals three times the induced drag, a condition associated with the maximum $\left(C_L^{1/2}/C_D\right)$ ratio. This condition is given by:

$$C_{D,0} = 3KC_L^2 \tag{6}$$

The corresponding velocity for this condition is:

$$V_{\left(C_L^{1/2}/C_D\right)_{\text{max}}} = \left(\frac{2}{\rho_\infty}\sqrt{\frac{3K}{C_{D,0}} \cdot \frac{W}{S}}\right)^{1/2} \tag{7}$$

As fuel is burned, the instantaneous rate of change of aircraft weight W is proportional to the required thrust T_R and the thrust-specific fuel consumption c_{tp} (Leishman, 2025):

$$\frac{dW}{dt} = T_R \, c_{tp} \tag{8}$$

Let W_0 be the weight at $t = t_0$ and $W_1 = W_0 - W_f$ the weight at $t = t_1$ after burning fuel mass W_f. Integrating ((8)) yields the flight time Δt:

$$\Delta t = \int_{t_0}^{t_1} dt = -\int_{W_0}^{W_1} \frac{dW}{c_{tp}\,T_R} = \int_{W_1}^{W_0} \frac{dW}{c_{tp}\,T_R} = \int_{W_0 - W_f}^{W_0} \frac{dW}{c_{tp}\,T_R} \tag{9}$$

If W_f is the total fuel onboard, then Δt equals the endurance E:

$$E = \int_{W_0 - W_f}^{W_0} \frac{dW}{c_{tp}\,T_R} \tag{10}$$

During steady, level flight ($T_R = D$, $L = W$), substitution into ((10)) gives

$$E = \int_{W_0 - W_f}^{W_0} \frac{1}{c_{tp}} \left(\frac{L}{D}\right) \frac{dW}{W} = \int_{W_0 - W_f}^{W_0} \frac{1}{c_{tp}} \left(\frac{C_L}{C_D}\right) \frac{dW}{W} \tag{11}$$

For a jet powered aircraft, endurance is proportional to C_L/C_D (Leishman, 2025). Assuming c_{tp} and C_L/C_D constant, ((11)) integrates to the Breguet endurance equation:

$$E = \frac{1}{c_{tp}} \left(\frac{C_L}{C_D}\right) \ln\!\left(\frac{W_0}{W_0 - W_f}\right) \tag{12}$$

For optimal endurance performance in a jet aircraft:

- Operate at or near the engine's minimum TSFC.
- Carry the maximum structural fuel load W_f.
- Fly at or near the best lift-to-drag ratio C_L/C_D.

The relation between range, endurance, and velocity is illustrated in Figure 2.4, which shows that maximum endurance is achieved at a lower velocity corresponding to maximum C_L/C_D, while maximum range occurs at a higher velocity where $C_L^{1/2}/C_D$ is maximised.

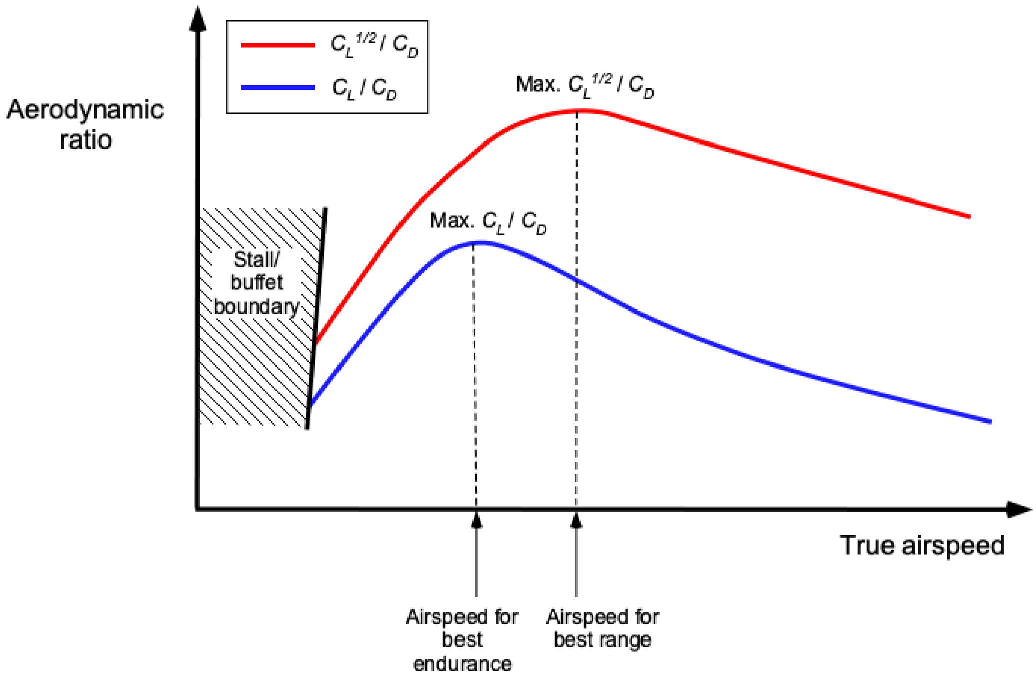


Fig. 2.4. Jet aircraft's Best Endurance and Best Range Airspeeds are Obtained at Different Velocities and Conditions (Leishman, 2025).

2.5.6 Specific Air Range and Endurance

During the cruise phase of flight—where altitude and velocity are approximately constant—aircraft efficiency is typically evaluated using point performance metrics. Two of the most vital are *Specific Air Range* (SAR) and *Specific Endurance* (SE), which quantify how efficiently an aircraft converts fuel into distance flown and time aloft, respectively (APD, 2024). These parameters are key to determining optimal cruise trajectories, whether to maximise range or endurance.

The specific air range is defined as the distance flown per unit of fuel weight consumed, similar to fuel economy in terrestrial vehicles. For a jet aircraft in steady, level flight, SAR can be expressed as:

$$\mathrm{SAR} = \frac{V}{T_R \cdot c_{tp}} \tag{13}$$

where V is the true airspeed, T_R is thrust required (equal to drag D in level flight), and c_{tp} is the thrust-specific fuel consumption (TSFC). As aircraft mass decreases during cruise, the required thrust decreases, thereby increasing SAR over time. Hence, SAR improves continuously with decreasing weight, assuming constant Mach number and altitude (APD, 2024).

The condition for maximum SAR does not coincide with minimum drag. Instead, SAR is maximised when the aircraft flies at the condition of maximum $C_L^{1/2}/C_D$, which balances drag and velocity to optimise fuel efficiency per unit distance. Operationally, SAR increases with cruise altitude and decreases with increasing Mach number. Thus, long-range cruise performance is typically achieved at higher altitudes and moderate speeds.

Specific endurance measures the time aloft per unit of fuel weight burned and is critical for missions requiring loiter or surveillance capability (APD, 2024). For a jet aircraft, it is given by:

$$\mathrm{SE} = \frac{1}{T \cdot c_{tp}} \tag{14}$$

This expression reveals that endurance is maximised by minimising both the required thrust and the TSFC. Unlike SAR, the optimal condition for SE occurs when the aircraft is flown at the velocity corresponding to maximum C_L/C_D, which prioritises aerodynamic efficiency over speed (APD, 2024).

In summary, SAR and SE are essential tools for quantifying an aircraft's cruise performance. SAR is used to determine fuel-optimal distance trajectories, while SE is applied in scenarios where flight duration is the primary constraint. Their difference is the foundation for differing mission design priorities, and their sensitivity to aerodynamic and propulsion parameters also corroborates flight at optimal conditions for fuel efficiency.

3 Methodology

The design and application of a performance analysis tool require a robust and iterative methodology, structured to align with each of the main project objectives and constraints. Given the mission's requirement to comply with UK CAA Open Category regulations, this design process prioritised a strict emphasis on aircraft weight and operational altitude limits. Figure 3.1 displays the general methodology workflow followed throughout the project.

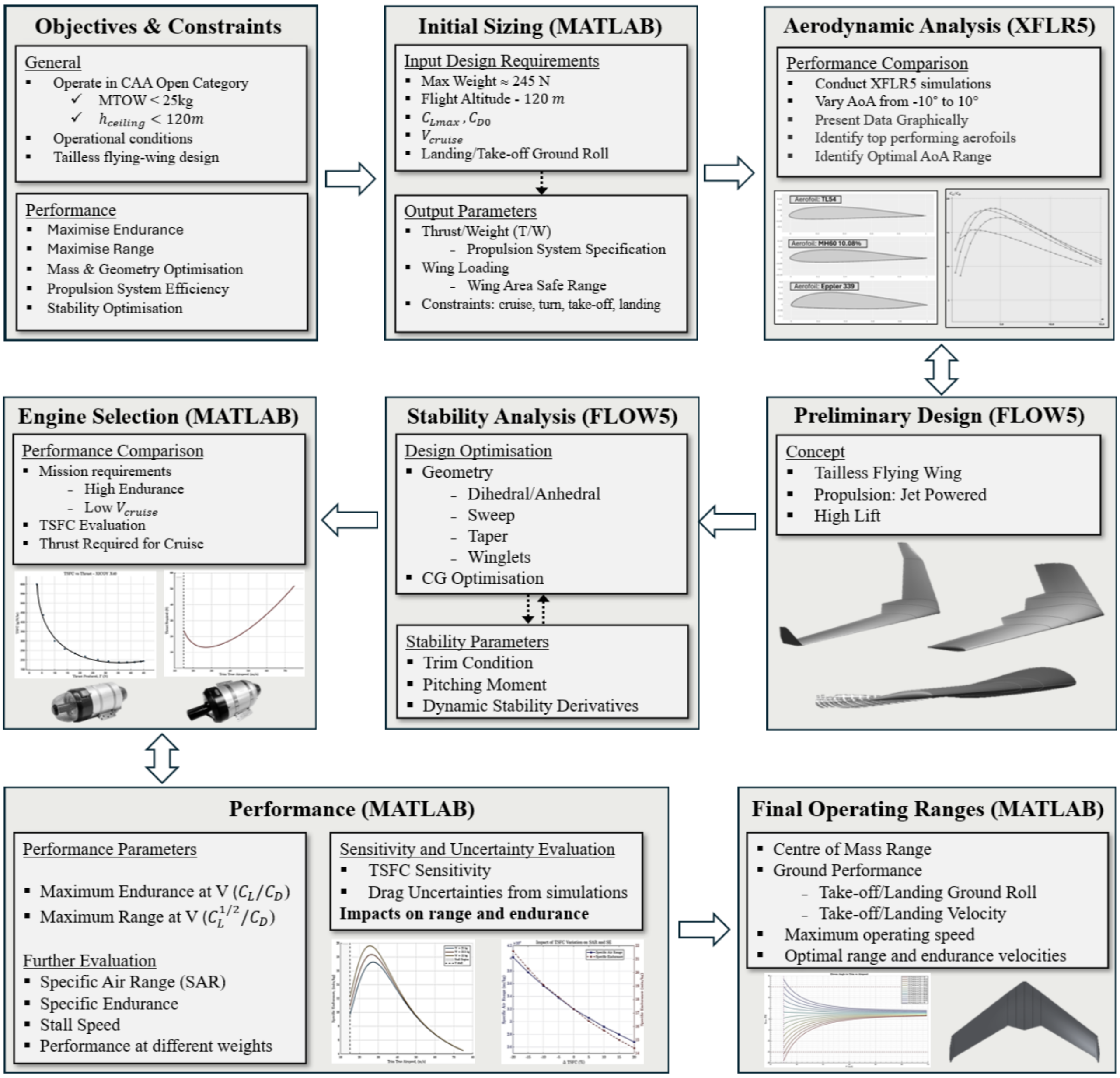


Fig. 3.1. Methodology Workflow

3.1 Defining the Requirements

The design process begins with defining both general and performance-related requirements. In accordance with CAA Open Category guidelines, the UAS must have a maximum take-off weight (MTOW) under 25 kg and a flight ceiling height below 120 meters ($\simeq$393 ft). When considering performance, the UAS aims to maximise endurance and range, which require careful selection of the aerofoils and the propulsion system to maximise the glide ratio and the range parameter $C_L^{1/2}/C_D$.

3.2 Initial Sizing

This phase involves conducting a literature review, looking into previous successful UAS designs and missions and tailoring the input parameters to match the requirements of this mission. These include cruise velocity, maximum lift coefficient (C_{Lmax}), zero lift drag coefficient (C_{D0}), and desired take-off and landing distances. The constraints diagram is implemented using MATLAB; Equations (15) through (18) are the key functions used to plot the constraints diagram such that $T/W = f(W/S)$, where T is thrust, W is weight, and S is the wing area. In this form, the wing loading (W/S) is plotted along the x-axis and the thrust-to-weight ratio (T/W) along the y-axis Gudmundsson, 2022). Design points can be selected from the bounded area which illustrates the feasible design space.

T/W for a Desired Take-off Ground Run Distance

$$\frac{T}{W} = \frac{1.21}{g\rho C_{L_{\max}} S_{TG}} \frac{W}{S} + \frac{0.605}{C_{L_{\max}}} \left(C_{D,\mathrm{TO}} - \mu C_{L,\mathrm{TO}}\right) + \mu \tag{15}$$

where,

$C_{D,\mathrm{TO}}$	Drag coefficient during take-off
$C_{L,\mathrm{TO}}$	Lift coefficient during take-off
$C_{L_{\max}}$	Max lift coefficient (take-off config)
μ	Ground friction constant
S_{TG}	Take-off ground roll [m]
g	Gravitational acceleration [m/s^2]

T/W for a Desired Rate of Climb

$$\frac{T}{W} = \frac{V_V}{V_\infty} + \frac{q}{\frac{W}{S}} C_{D_{\min}} + \frac{k}{q}\frac{W}{S} \tag{16}$$

where,

V_∞	Airspeed [m/s]
V_V	Vertical speed [m/s]
q	Dynamic pressure
$C_{D_{\min}}$	Minimum drag coefficient
k	Induced drag factor

T/W for a Level Constant Velocity Turn

$$\frac{T}{W} = q\left[\frac{C_{D_{\min}}}{\frac{W}{S}} + k\left(\frac{n}{q}\right)^2 \frac{W}{S}\right] \tag{17}$$

where,

n	Load factor $= \frac{1}{\cos(\theta)}$
k	Lift-induced drag constant

T/W for a Target Total Landing Distance

$$\begin{aligned} S_{\mathrm{LDG}} = & 19.1h + \left[0.008 + 1.556\tau\sqrt{\frac{\rho C_{L_{\max}}}{\frac{W}{S}}} \right. \\ & \left. + \frac{1.21}{g}\left[\frac{0.605}{C_{L_{\max}}}\left(C_{D,\mathrm{LDG}} - \mu C_{L,\mathrm{LDG}}\right) + \mu\right]\right]\frac{\frac{W}{S}}{\rho C_{L_{\max}}} \end{aligned} \tag{18}$$

where,

h	Obstacle height
ρ	Air density at landing altitude
$C_{L_{\max}}$	Max lift coefficient (landing config)
$C_{L,\mathrm{LDG}}$	Lift coefficient during ground roll
$C_{D,\mathrm{LDG}}$	Drag coefficient during ground roll
μ	Ground friction (typically 0.3)
τ	Free-roll time before braking

3.3 Aerodynamic Analysis

To identify the most suitable aerofoils for a tailless flying wing configuration, a selection of candidate profiles, primarily reflexed aerofoils, are analysed using XFLR5 software. The aerofoil shape data points are available to download from websites such as 'Airfoil Tools'. These are further imported to the software where aerodynamic modelling simulations are conducted across a range of angles of attack and Reynolds numbers. This enables a comparative analysis of performance parameters such as glide ratio and maximum range plots.

The aerofoils with the most desirable aerodynamic characteristics are selected for 3D analysis on a simple flying wing configuration. For this project, all 3D modelling is conducted on FLOW5 software, although this is still possible in XFLR5. The results are tabulated and scored based on the most important factors the design requires, such as high lift-to-drag ratios. The analysis uses a Vortex Lattice Method (VLM) to predict their aerodynamic performance in a three-dimensional case, noting that this method underestimates the total drag by roughly 20%. The final best scoring aerofoils are chosen to make up the flying wing profile.

3.4 Stability Analysis

This stage focuses on optimising the centre of mass (CoM) and wing geometry to ensure static and dynamic stability. Weight and balance stage is a particular important sector in a flying wing configuration due to the absence of a conventional fuselage and tail. Design aspects such as pitching moments, trim condition and dynamic stability derivatives are significant aspects that will need to be fine-tuned to allow effective control and manoeuvring.

FLOW5 software has the ability to calculate the stability derivatives, which indicate whether the aircraft is stable or not based on their positive or negative sign. The tool is also able to compute the dynamic modes, such as the phugoid, short period, roll, and yaw. The damping coefficient and time-to-half are important elements that require fine-tuning to find the optimum points that favour all modes. Conducting this analysis will likely lead to design changes, such as the addition of winglets, dihedral, geometric twist, and adjustments to the taper ratio.

3.5 Performance Modelling

3.5.1 Propulsion System Selection

The engine selection process is guided primarily by the constraint analysis, which determined the required thrust output to meet mission performance criteria. Selected engines are evaluated based on their ability to deliver the necessary thrust for each mission segment while optimising for fuel efficiency. Thrust-specific fuel consumption (TSFC) is a key parameter in the evaluation, aiming to achieve maximum endurance and range with the lowest possible fuel burn rate.

3.5.2 Specific Air Range and Endurance

For the finalised, stable aircraft design and selected propulsion system, the Specific Air Range (SAR) and Specific Endurance (SE) calculations can be implemented in MATLAB as follows below (APD, 2024).

Performance calculations:

Aerodynamic Coefficients Calculation:

$$C_L = \frac{W}{0.5 \times \rho \times V_{\text{trim}}^2 \times S}$$
$$C_D = C_{D0} + K \times C_L^2$$

Thrust Required:

$$T_R = 0.5 \times \rho \times V_{\text{trim}}^2 \times S \times C_D$$

Performance Metrics:

$$\text{SAR} = \frac{V_{\text{trim}}}{T_R \times c_{tp}} \quad \text{[meters flown / kilogram of fuel]}$$
$$\text{SE} = \frac{1}{T_R \times c_{tp}} \times \frac{1}{60} \quad \text{[minutes flown / kilogram of fuel]}$$

3.5.3 Code Implementation Notes

- The code structure is adjusted to calculate performance across a variety of altitude and weights.
- All aerodynamic parameters (C_{D0}, K, S) use the finalised design values
- The propulsion parameter c_{tp} (thrust-specific fuel consumption) corresponds to the selected engine with units kg/N/s
- Results can be visualised as:
  - SAR vs. Airspeed at different altitude/weights for range performance analysis
  - SE vs. Airspeed at different altitude/weights for endurance evaluation

The overall performance analysis sector focuses on evaluating the most optimal velocities to travel for maximum range and endurance. A further sensitivity analysis becomes especially important in flying wing designs to ensure performance remains robust against changes in drag and TSFC, as these parameters may vary in different conditions and impact mission effectiveness.

3.5.4 Finalised Design Operating Ranges

Finally, a concluding verification section is included to outline all performance results and operating ranges for the final design. The 'Flight Dynamics' MATLAB tool is able to take design inputs for a tailless configuration and display the stable range of center of mass (CoM) locations, illustrated by the positive gradient curves within, which represent the range of elevator deflection angle (Flight Dynamics, 2025).

Furthermore, the 'Take-off Performance' and 'Landing Performance' tools follow a similar method of taking design inputs and outputting performance metrics such as take-off/landing velocity, take-off/landing distance, as well as braking distance.

4 Results and Analysis

4.1 Introduction

This section presents the results and key finding from the iterative design and analysis process, covering aerodynamic, stability, and performance aspects of the proposed jet-powered tailless UAS. It begins with the initial constraint analysis which guided the narrowing of feasible design options.

Aerofoil selection, wing configuration, and stability were assessed using 3D FLOW5 simulation and low fidelity XFLR5 methods, while propulsion and mission performance were assessed through analytical models custom developed MATLAB tools.

By combining these elements, a complete understanding of the aircraft's capabilities is achieved, with the final design's operating ranges evaluated against criteria for range, endurance, takeoff and landing to confirm its suitability for high efficiency demanding missions.

4.2 Constraints Analysis

The constraint analysis is the process that conducts the initial sizing of the aircraft. This is one of the first tasks in any new aircraft design, which allows the aircraft's required wing area and installed thrust to be assessed such that it meets all performance requirements and desired values.

Figure 4.1 illustrates the constraint diagram created in MATLAB, showcasing the constraints within the design space. The constraints are represented using thrust loading (T/W) as a function of wing loading (W/S). The white region in Figure 4.1 is the set of acceptable solutions, while the shaded regions represent the impracticable solutions. Points B and C represent the optimal design solutions, as they achieve all design requirements with the minimum necessary thrust loading. Design A exceeds all; however, E exceeds the maximum landing distance, and point D does not meet the design requirements.

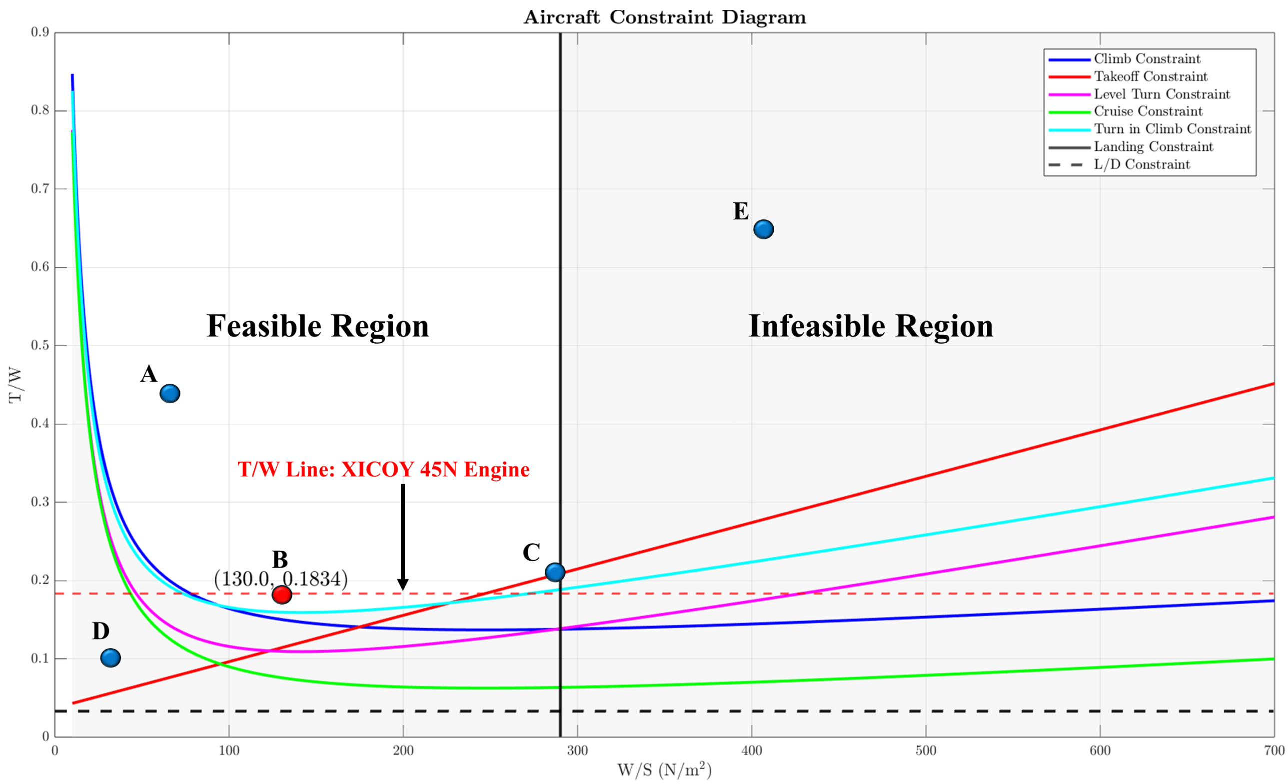


Fig. 4.1. Constraint Diagram (Thrust to Weight (T/W) vs Wing Loading W/S).

The aircraft's total mass was constrained to under 25 kg to comply with CAA open category operations. This mass limitation, along with the design input parameters in Table 4.1, such as the cruise velocity of 30 m/s and $C_{L_{\max}}$ of 1.2, guided the constraint analysis. Examination of the constraint diagram revealed a minimum feasible thrust-to-weight ratio (T/W) of 0.159. Based on this value and the specified MTOW, the design outputs include a minimum required thrust of 39 N and a wing area range of 1.23-2.45 m^2, as shown in Table 4.2.

Input Parameter	Value	Unit
Take-off Weight	245.25	N
Ceiling Altitude	120	m
Cruise Velocity	30	m/s
Landing Ground Roll	250	m
C_{D0}	0.014	–
K	0.07	–
$C_{L_{\max}}$	1.2	–

Table 4.1. Design Input Parameters

Output Parameter	Value	Unit
T/W	0.183	–
Safe W/S Margin	100-200	N/m^2
Wing Area Range (S)	1.23-2.45	m^2
Installed Thrust	>39	N

Table 4.2. Design Output Parameters

A challenge encountered when using this method is that, since the aircraft's geometry is unknown, parameters such as $C_{D_{\min}}$, $C_{D_{Take-off}}$, $C_{L_{Take-off}}$, and k are also unknown. To address this, I referred to existing aircraft within the same class as the one being designed. Table B.1 and B.2 illustrate a range of typical values, serving as an aid to peruse this design.

4.3 Aerofoil Selection Process

The aerofoil design plays a vital role in determining lift and drag, directly influencing overall aircraft performance. In tailless flying wing configurations, aerofoil selection is particularly of importance, as it must support both lift generation and longitudinal stability. To achieve longitudinal stability in the absence of a conventional tail, the flying wing design should employ reflexed aerofoils to counteract inherent nose-down pitching moments.

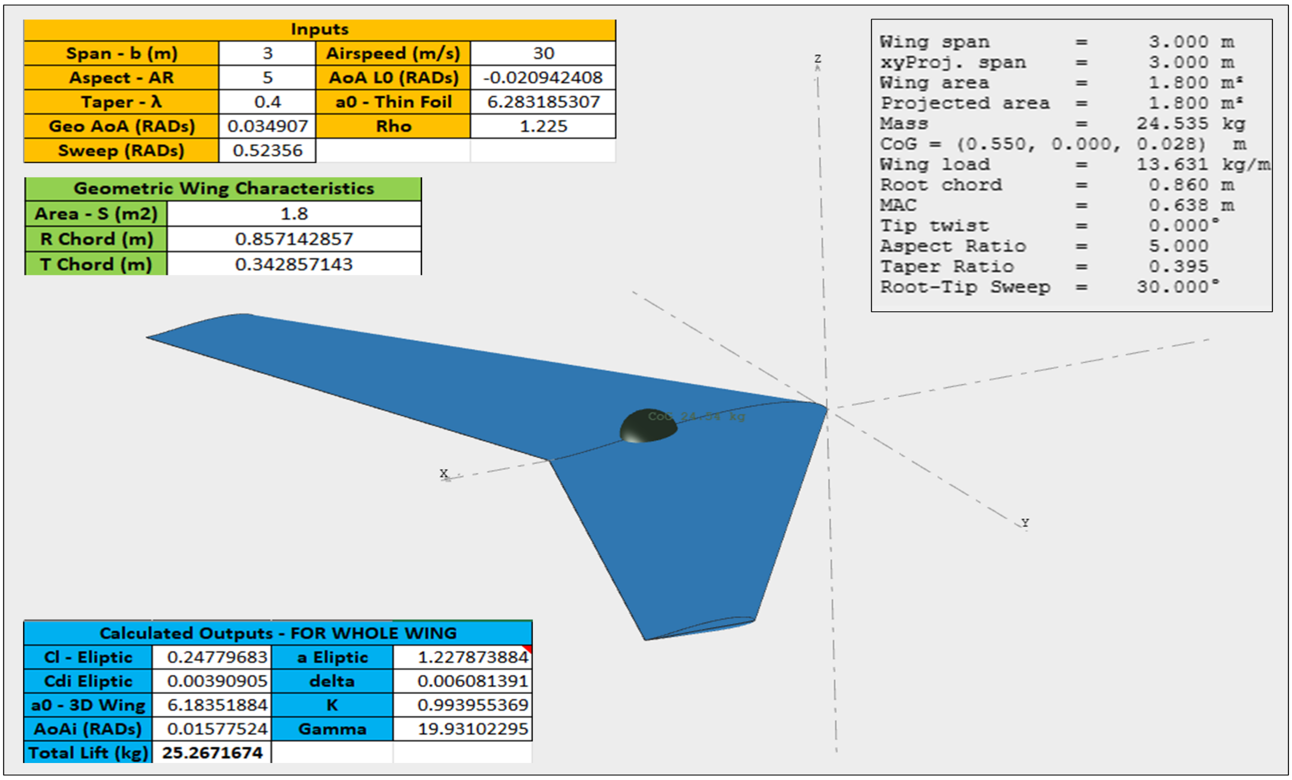


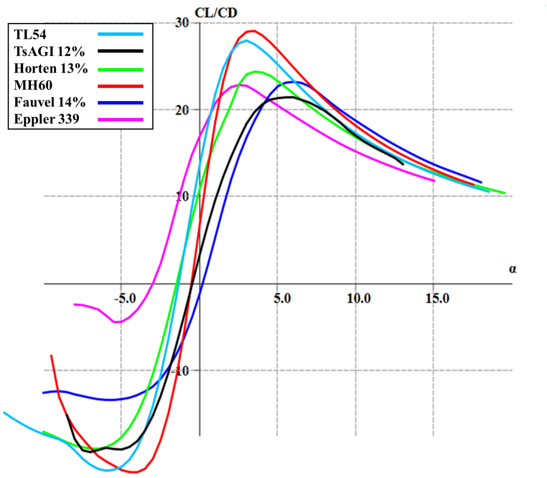
Fig. 4.2. Simple Flying Wing Design and Parameters Used in the Analysis

Figure 4.2 shows the simple flying design created in FLOW5 which assists the selection of the aerofoil. Its shape and parameters were calculated using an Excel wing geometry tool developed by Kushal Singh (Singh, 2025). The aircraft's basic parameters, derived from the constraints analysis in Section 4.2, were used in the tool to ensure the wing generated the target lift of 25 kg. This ensured the wing design met the lift requirement so making the analysis representative of a practical configuration capable of generating sufficient lift for flight. For the analysis, the wing geometry and all parameters remain identical; only the aerofoil sections vary along the span to compare their aerodynamic performance.

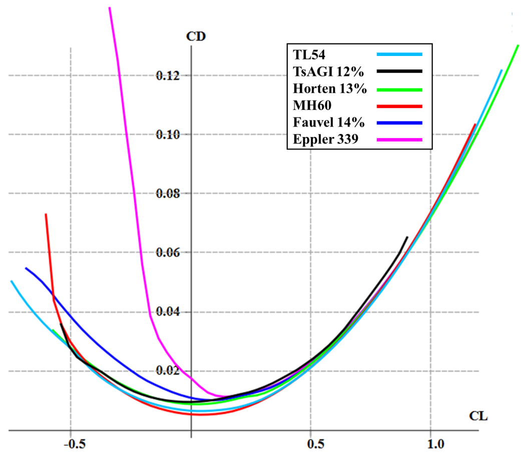
Parameter	Value
Average Reynolds Number (Cruise)	1,000,000
Average Reynolds Number (Stall)	400,000
Cruise Velocity [m/s]	30
Target $C_{L_{\max}}$	1.2
Target Cruise C_L	0.2
Target ROC C_L	0.4

Table 4.3. Reynolds Numbers and Design Parameters Used for the Analysis

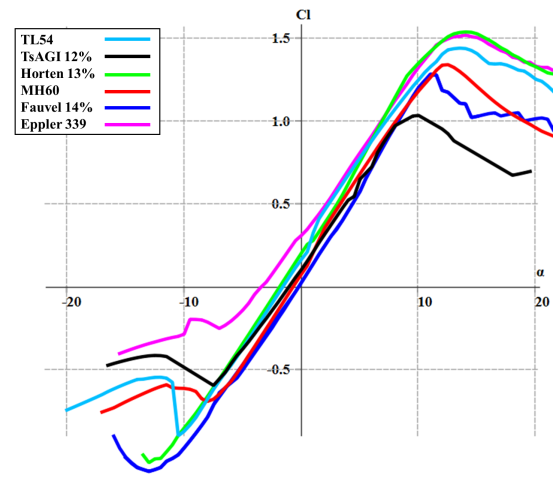
The analysis involved conducting a Type 1 Uniform Triangle Panel method, where the velocity is fixed at 30 m/s, to compute key performance metrics including lift, drag, and pitching moment coefficients. Table 4.3 outlines the main parameters and Reynolds numbers used throughout the simulations.



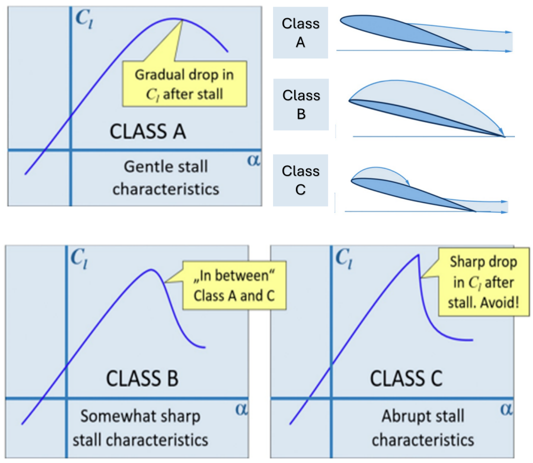
(a) Lift-to-drag ratio vs. Angle of Attack.



(b) (C_D vs. C_L) Illustrating The Drag Bucket.



(c) C_L vs. Angle of Attack.



(d) Aerofoil Stall Characteristics (Gudmundsson, 2022).

Fig. 4.3. Aerofoil Selection Comparison: Lift-to-Drag, Drag Variation, Lift vs Angle of Attack, and Stall Behaviour.

The analysis focused on how each aerofoil performs under consistent flow conditions, allowing for direct comparison. The results from this analysis are shown in Figure 4.3, providing a visual comparison of aerodynamic efficiency, stall behaviour, and overall performance trends.

Parameter	**FW1**	**FW2**	**FW3**	**FW4**	**FW5**	**FW6**
Aerofoil Used	TL54	TsAGI 12%	Horten 13%	MH60	Fauvel 14%	Eppler 339
Aerofoil Thickness Ratio	0.1	0.12	0.13	0.1	0.14	0.135
Optimal AoA [°]	3–4.5	5–7.5	3.5–4.5	3–5	5–7	2.5–4
Best Glide Ratio	28.1	21.4	24.3	29.1	23.3	22.8
Stall Class (A/B/C)	A	B	A	A	B	A
AoA for $C_{\mathrm{L}} = 0$ [°]	-1.2	-2.0	-2.3	-0.31	0	-3
Cruise C_{m} at CL ≈ 0	-0.0171	-0.005	-0.0008	-0.003	0.030	-0.040
C_{Dmin}	0.007	0.010	0.009	0.005	0.010	0.013
C_{L} drag bucket start	0.12	0.28	0.22	0.10	0.17	0.30
C_{L} drag bucket end	0.57	0.58	0.60	0.55	0.50	0.70
$C_{\mathrm{L,cruise}}$ in bucket?	Yes	No	No	Yes	Yes	No
$C_{\mathrm{L,ROC}}$ in bucket?	Yes	Yes	Yes	Yes	Yes	Yes

Table 4.4. Aerodynamic Parameters for each Simple Flying Wing Design.

To narrow down the most suitable aerofoil, key performance metrics from the FLOW5 analysis were tabulated and compared across each flying wing configuration, as shown in Table 4.4. These include aerodynamic characteristics such as glide ratio, stall behaviour, pitching moment coefficient, and whether cruise and climb lift coefficients fall within the 'drag bucket'.

Scored Parameter	**FW1**	**FW2**	**FW3**	**FW4**	**FW5**	**FW6**
Glide Ratio	0.87	0.00	0.38	1.00	0.25	0.19
C_{Dmin}	0.75	0.375	0.5	1.00	0.375	0.00
Cruise C_{m}	0.61	0.90	1.00	0.98	0.41	0.00
$\mathrm{CL}_{\mathrm{cruise}}$ in bucket	1.0	0.0	0.0	1.0	1.0	0.0
$\mathrm{CL}_{\mathrm{ROC}}$ in bucket	1.0	1.0	1.0	1.0	1.0	0.0
Stall Class	1.0	0.5	1.0	1.0	0.5	1.0
AoA for CL=0	0.93	0.95	0.30	1.00	0.73	0.00
Average Score	**0.88**	**0.53**	**0.60**	**0.99**	**0.61**	**0.31**

Table 4.5. Aerofoil Performance Scores (1 Indicates the Highest Score in each Category, 0 the Lowest).

Furthermore, a simple scoring matrix is shown in Table 4.5, where each aerofoil performance parameter has been rated between 0 (worst) and 1 (best) in every category. These scores were weighted according to their relative importance to the design goals of a tailless, flying wing optimised for endurance and range, with greater the priority given to factors such as the lift-to-drag ratio and stability, which are especially significant to the purpose of this mission. The total averaged scores show that the MH60 aerofoil (Flying Wing 4) is dominant across the majority performance metrics, with the TL54 (Flying Wing 1) following closely in second place overall. In practice, additional design modifications are likely to occur, particularly as the aerofoil section will vary along the span. For instance, the Fauvel aerofoil (Flying wing 5) of 0.14 thickness ratio can be used in the root section of the wing to contain internal systems and act as a fuselage structure (further discussed in Section 4.4), and thinner sections may be used outboard to reduce drag and improve aerodynamic efficiency.

On the basis of the performance parameters considered, the MH60 aerofoil is selected for the main span as it carries the greatest lift-to-drag ratio, favourable stability due to its minimal pitching moment and gentle stall characteristics. The TL54 won't be considered as it has a larger drag and pitching moment coefficient worsening both static and dynamic stability in our tailless design. The TsAGI 12% is also excluded due to exhibiting the lowest lift-to-drag ratio among the candidates and its slightly harsher stall behaviour. For the root sections, three profiles with higher thickness - Fauvel 14%, Eppler 339, and Horten 13%w - ill be progressed for complete analysis in the next section. The Eppler 339's smooth stall characteristics and negative zero-lift angle of attack that will generate a stabilising nose-down pitching moment; the Fauvel 14% possesses a high glide ratio and strong stability characteristics to go along with its high overall score; and the Horten 13% has an extremely low coefficient of moment at zero lift, a relatively high negative zero-lift angle of attack along with the highest lift-to-drag ratio out of the three. The root aerofoils require further examination to find the most efficient profile to accompany the MH60 and house internal components without compromising aerodynamic efficiency.

4.4 Preliminary Wing Design

This section presents the initial generation of the flying wing design, based on the sizing constraints and aerofoil analyses presented earlier. In order to provide room for internal systems such as the avionics and engine, it is important to vary the aerofoil profile along the span, with a thicker section at the root and a thinner profile outboard. Initially, each of the thick root section were analysed separately, with either the Horten 13%, Fauvel 14%, or Eppler 339 aerofoil used at the root, and the MH60 profile employed across the rest of the wing.

4.4.1 Single Root Aerofoil Configuration

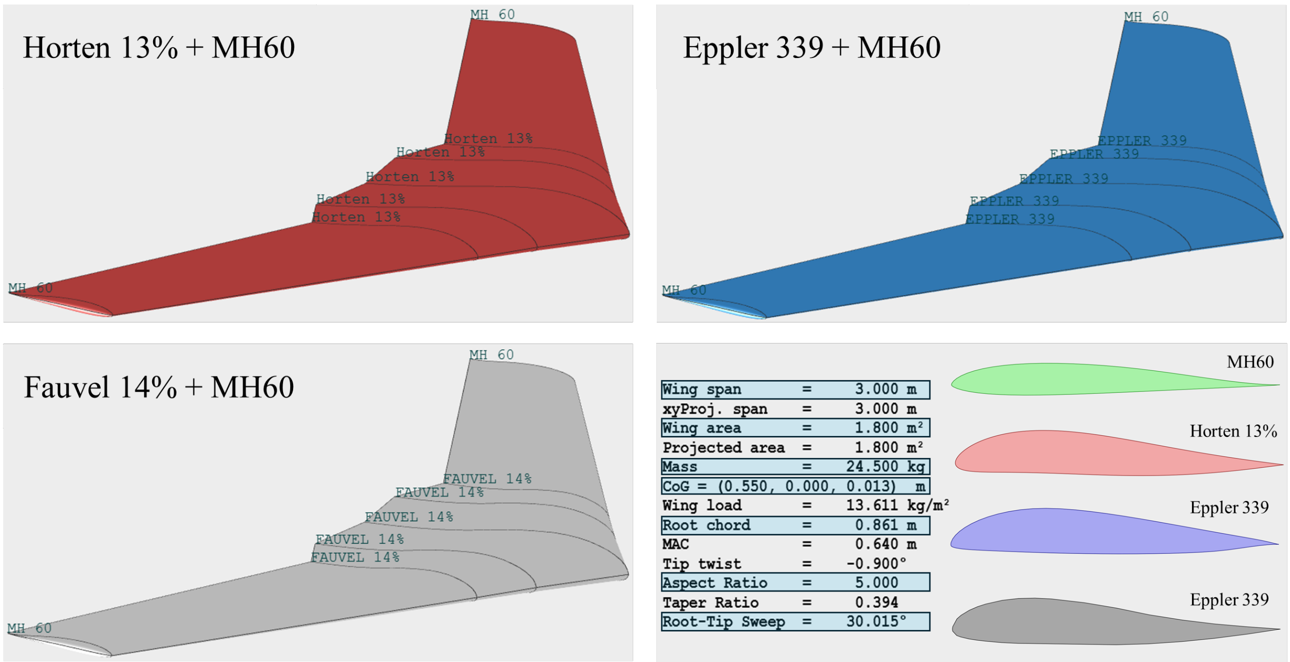
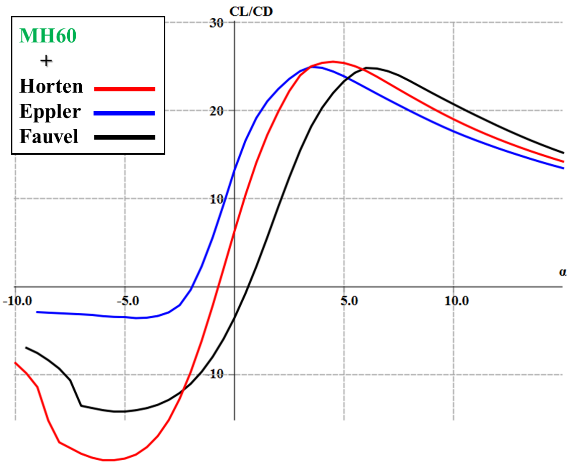
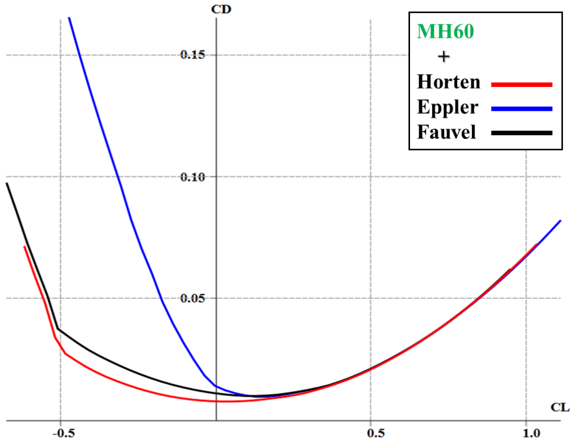


Fig. 4.4. Flying Wing Configurations With Different Root Aerofoils and MH60 Used Outboard.

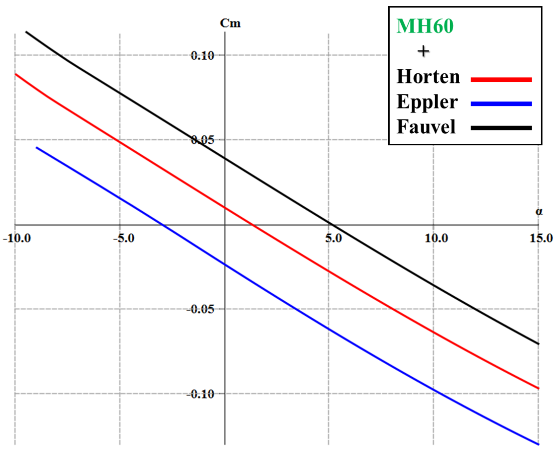
These three root aerofoil combinations were tested using FLOW5 ensuring the geometry, centre of mass location, flight speed of 30 m/s are kept constant. As illustrated in Figure 4.5, the Horten 13% with MH60 combination is the most efficient in cruise with a maximum L/D of 25.6 at an angle of attack of 4.5°. The Eppler 339 + MH60 configuration achieves a comparable maximum L/D of 25.0 at 3.5°, but it exhibits a significant spike in drag at negative C_L values compared with the other combinations. Moreover, its pitching moment profile crosses $C_\mathrm{m} = 0$ at a negative lift coefficient meaning it cannot trim at the positive lift required for level flight and is therefore unacceptable without requiring a redesign. The Fauvel 14% + MH60 configuration has the lowest peak L/D of 24.8 at 6°.



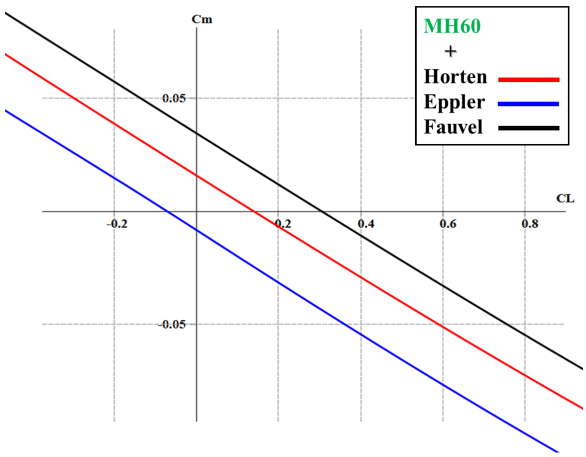
(a) Lift-to-Drag Ratio vs. Angle of Attack.



(b) Drag Variation (C_{L} vs. C_{D}).



(c) C_{m} vs. Angle of Attack.



(d) C_{m} vs. C_{L}.

Fig. 4.5. Performance Plots of the Flying Wing With Different Root Aerofoils Alongside the MH60 Outboard.

In summary, when varying the root aerofoils in such combinations, the Horten 13% alongside the MH60 delivers the highest L/D and a practical trim solution, outperforming both the Eppler 339 and Fauvel 14%.

4.4.2 Multiple Root Aerofoil Configuration

The next phase of the design investigates the performance of various combinations involving all three candidate root aerofoils, i.e. Eppler 339 (E), Horten 13% (H), and Fauvel 14% (F), in combination with the MH60 (M) profile along the outer span. This process aims to evaluate the potential benefits of combining different root aerofoil sections, rather than relying on a single root profile. As illustrated in Figure 4.6, four stable configurations were chosen for analysis, each utilising a unique combination of the E, H, F to go alongside the MH60 profile, investigating their effect on aerodynamic performance, stability, and trim capability.

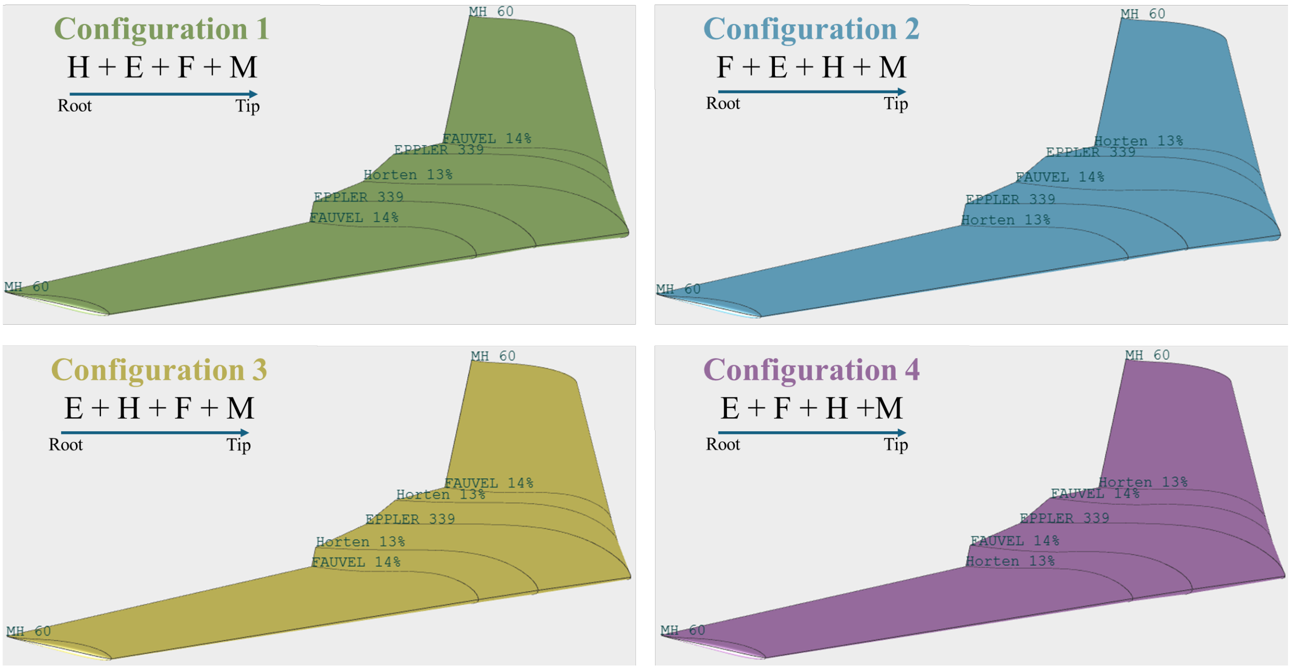
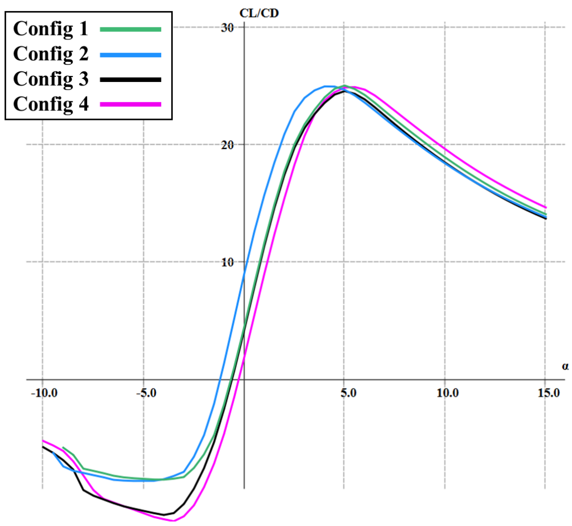
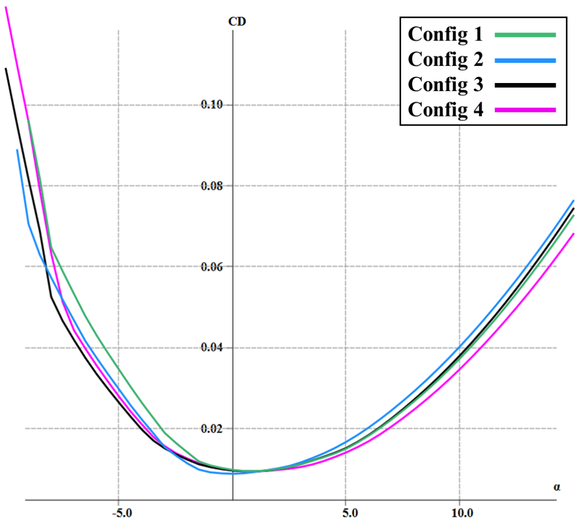


Fig. 4.6. Flying Wing Configurations Using Combinations of Eppler 339, Horten 13%, and Fauvel 14% at the Root, With MH60 Used Outboard.

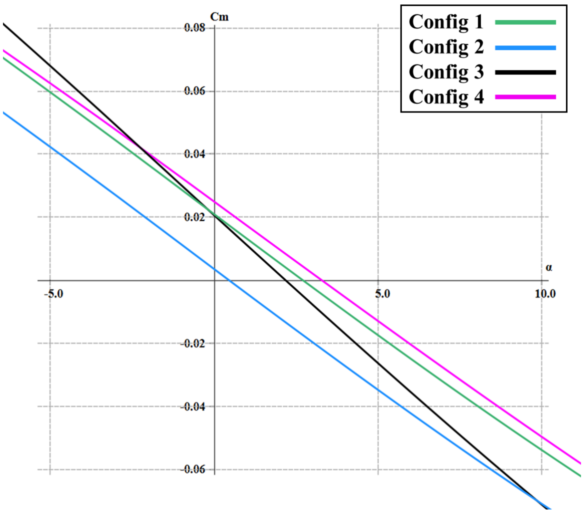
The results for this analysis are shown in Figure 4.7 and the different root aerofoil combinations demonstrate near- identical lift-to-drag performance, with only minor variations across the set. Configuration 2 achieves the highest peak L/D ratio of 25.12 at an angle of attack of 4°, followed closely by Configurations 1 and 4, each reaching approximately 25.0 at 5° angle of attack. Configuration 3 performs slightly worse than the rest, peaking at 24.6 at 5° angle of attack. While the L/D values are close, the primary differences arise in their trim behaviour. All configurations are statically stable, indicated by a negative slope in the C_m–C_L plots. Configuration 1 trims at 2.7° with a corresponding C_L of 0.22; Configuration 2 trims significantly earlier at just 0.45° with C_L of 0.11. Configuration 3 trims at 2.18° with and a C_L of 0.19, while Configuration 4 trims at 3.4° and C_L of 0.23. These values are calculated using a consistent centre of mass location, fixed to provide a static margin of 10% across all configurations, in order to allow a fair comparison. A more detailed stability analysis considering variations in CoM position and their effect on trim behaviour is presented in Section 4.5. From the drag polar plots, the minimum drag coefficients are nearly identical for all cases; however, Configuration 1 slightly more drag at negative lift coefficients compared to others.



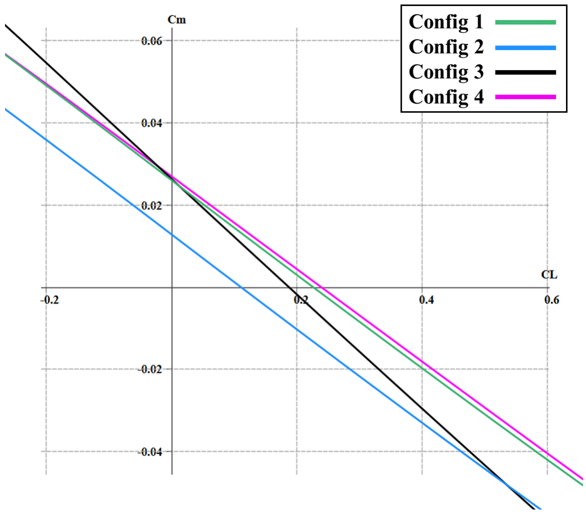
(a) Lift-to-Drag Ratio vs. Angle of Attack.



(b) Drag Variations C_D vs. Angle of Attack.



(c) Lift Coefficient C_L vs. Pitching Moment C_m



(d) C_m vs Angle of Attack.

Fig. 4.7. Aerodynamic Performance Plots of the Flying Wing With Different Configurations of Multiple Root Aerofoils.

Although all four root aerofoil combinations deliver similar lift-to-drag performance, Configuration 4 provides the most balanced outcome, with an acceptable L/D ratio and practical trim point. Configurations 2 and 4 show favourable aerodynamic characteristics, while Configuration 3 remains a strong candidate, despite having the lowest lift-to drag ratio. Overall, all configurations demonstrated viable aerodynamic performance. Configuration 4 stood out the most for its balanced overall efficiency combined with stable trim behaviour. However, this analysis suggests that the choice of root aerofoil has a minor impact, and any of the tested configurations could be practical with minimal complications. Instead, the key determinant is the dynamic stability modes, which define the suitability and controllability of each configuration.

4.5 Stability Analysis

The stability analysis of a tailless flying-wing configuration is one of the most significant parameters to analyse; the absence of a conventional tailplane means that longitudinal and lateral stability must be designed into the wing itself. To develop a stable arrangement, I have iteratively modified the wing planform and design by adjusting parameters such as sweep angle, dihedral, washout, and further implementing a winglet and adjusting the centre of mass location. The following section focuses on optimising the dynamic stability, detailing how each of the geometric parameters has been altered, showing their effect on stability derivatives and modal damping ratios, and describing how the final design meets both static and dynamic stability requirements for a stable, tailless UAS.

Derivative	Description	Notes	Desired Value
$C_{Y\beta}$	Side force vs. sideslip (β)	Lateral-directional stability	< 0
C_{Yp}	Side force vs. roll rate (p)	Rolling response	< 0 (close to 0)
C_{Yr}	Side force vs. yaw rate (r)	Positive yaw response	> 0
$C_{l\beta}$	Roll moment vs. sideslip (β)	Restores rolling stability	< 0
C_{lp}	Roll moment vs. roll rate (p)	Roll damping	< 0
C_{lr}	Roll moment vs. yaw rate (r)	Coupling effect	< 0 (can be > 0)
C_{ma}	Pitch moment due to AoA (α)	Longitudinal static stability	< 0
C_{mq}	Pitch moment vs. pitch rate (q)	Pitch damping	< 0
$C_{n\beta}$	Yaw moment vs. sideslip (β)	Directional stability	> 0
C_{np}	Yaw moment vs. roll rate (p)	Yawing tendency	< 0 (can be > 0)
C_{nr}	Yaw moment vs. yaw rate (r)	Yaw damping	< 0

Table 4.6. Aerodynamic Stability Derivatives with Desired Values

Table 4.6 highlights the key non-dimensional stability derivatives alongside their desired signs, such as $C_{l_p} < 0$ for roll damping and $C_{n_\beta} > 0$ for weathercock stability, which enable the aircraft to return to its equilibrium condition with minimal oscillations following a disturbance.

To achieve a dynamically stable design, it is important to understand how the wing's geometry influences the stability derivatives, as shown in Table 4.7. For instance, wing dihedral primarily controls the value and sign of C_{l_β}, geometric twist and the centre of mass location determine C_{m_α}, and winglets enhance both C_{Y_r} and C_{n_β}. By cross-referencing this with the target values in Table 4.6, iterative adjustments to parameters such as sweep, twist, dihedral, taper ratio, and the inclusion of winglets effectively steered the derivatives toward the desired values and signs.

Derivative	Geometric Contributors
$C_{Y\beta}$	Winglets/Sweep/Taper
C_{Yp}	Wings Geometry/ Dihedral/AR
C_{Yr}	Wing Geometry/Winglets
$C_{l\beta}$	Wing Dihedral
C_{lp}	Wing Geometry
C_{lr}	Wing Dihedral/Sweep
C_{ma}	Wing Twist/CoM Position
C_{mq}	Wing Sweep
$C_{n\beta}$	Wing Twist/Sweep/Taper
C_{np}	Wing Geometry
C_{nr}	Wing Twist/Sweep/Taper

Table 4.7. Aerodynamic Derivatives and Their Primary Geometric Contributors.

Initial refinements from Design 0 (Section 4.4.2) involved increasing the taper ratio, shifting the CoM, and adding extra geometric twist as displayed in Figure 4.8. These changes helped strengthen pitch stability, driving $C_{m\alpha}$ more negative (–0.533 to –0.592) and boosting pitch-rate damping (C_{mq} –1.574 to –1.867). Roll damping (C_{lp}) improved slightly, while weathercock stability (C_{n_β}) remained marginally positive. The short-period mode softened slightly in frequency (0.072 to 0.065 Hz) with unchanged damping, while the phugoid sped up to 1.97 Hz. Roll damping improved moderately, but Dutch-roll remained the main issue, as its damping ratio is negative—indicating unresolved yaw–roll coupling.

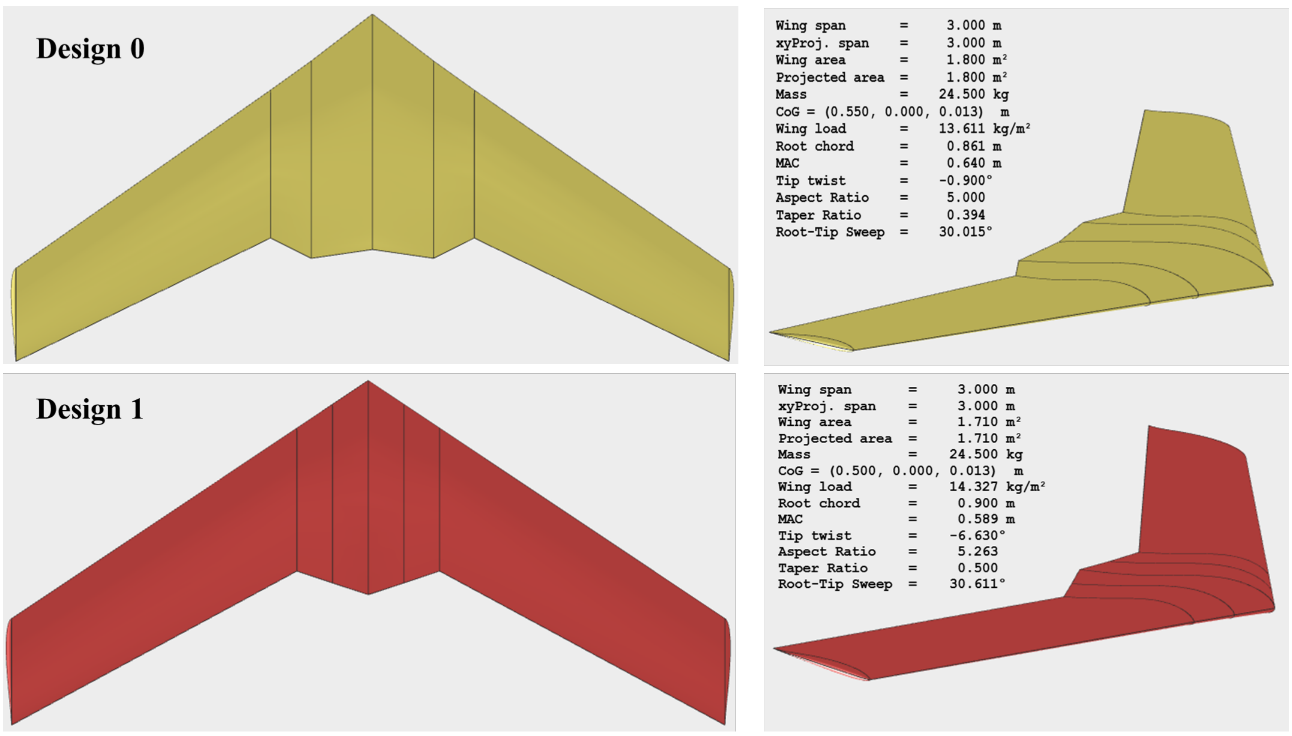


Fig. 4.8. Modification of Design 0 to Design 1 Showing Geometric Changes to Improve Stability.

Following the modifications, winglets were added along with a slight increase in sweep, necessitating a small CoM adjustment illustrated in Figure 4.9. Static longitudinal stability weakened slightly ($C_{m\alpha}$ from –0.592 to –0.540) but remained comfortably within acceptable limits. Lateral stability improved, with a more negative $C_{l\beta}$ (–0.061 to –0.079), and directional stability ($C_{n\beta}$) more than doubled (+0.0022 to +0.0052). These changes significantly enhanced Dutch-roll characteristics and weathercock response. The Dutch-roll mode stabilised, now exhibiting a positive damping ratio ($\zeta = 0.020$), marking a key benefit of the winglets. The spiral mode also shows marked improvement, as the half-life of the disturbance amplitude decreases substantially, demonstrating increased stability.

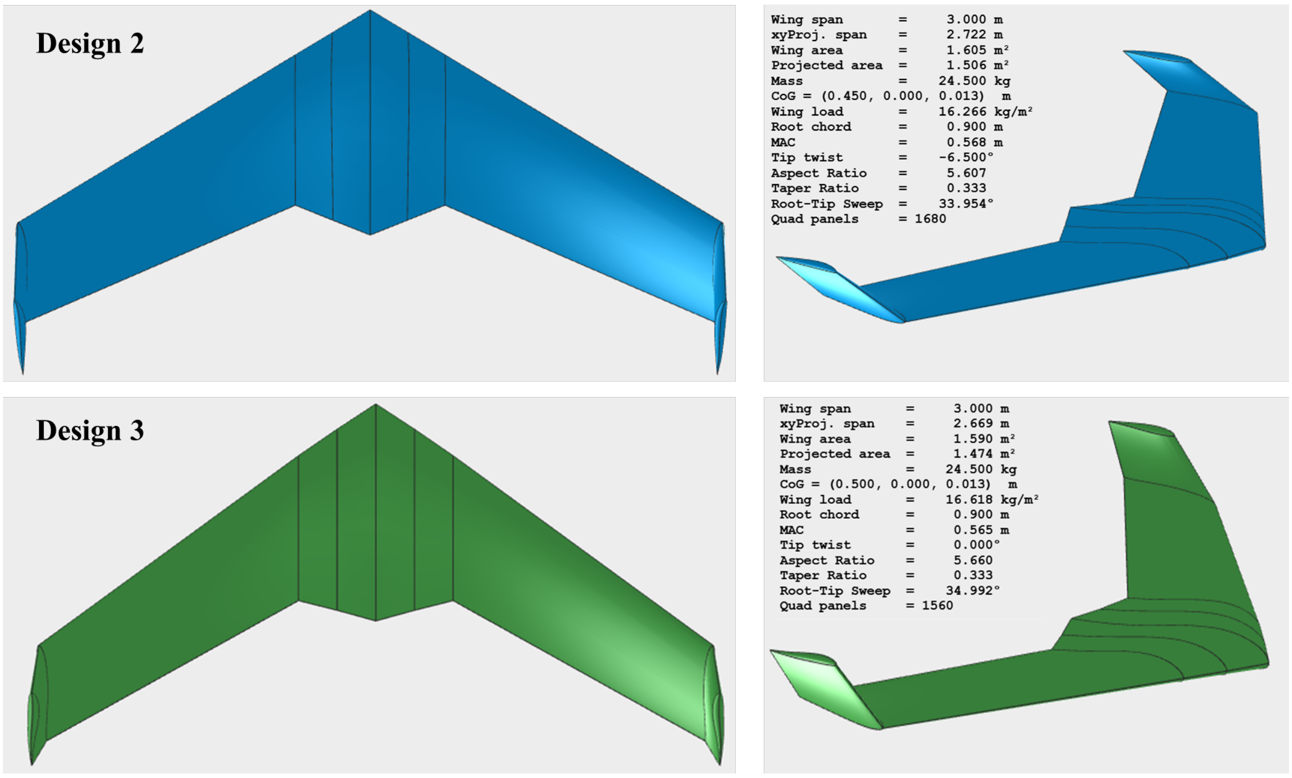


Fig. 4.9. Implementation of Winglets and Anhedral to Improve Dynamic Stability.

Final changes introduced 5° anhedral and a cambered winglet aerofoil (Fauvel 14%), further tuning lateral stability and enhancing lift. Yaw related derivatives $C_{Y\beta}$, C_{Yr}, and C_{n_β} improved significantly, indicating a stronger directional response. In particular, C_{n_β} rose sharply from 0.0052 to 0.0383, and Dutch-roll stability noticeably increased (= 0.050). Spiral-mode decay also quickened, with its half-life falling to under 14 s. Pitch damping (C_{mq} from –1.70 to –2.06) and static stability ($C_{m\alpha}$ from –0.540 to –0.665) both strengthened, and short-period damping improved, yielding a smoother yet well-damped longitudinal response. Overall, the final Design 3 (Figure 4.9) achieves the desired balance of dynamic stability and control.

Derivative	Design 0	Design 1	Design 2	Design 3
C_{Xu}	-0.009692	-0.007136	-0.006730	-0.005151
C_{Xa}	0.133602	0.128407	0.120216	0.103510
$C_{Y\beta}$	0.000159	-0.000822	-0.038987	-0.169113
C_{Yp}	-0.038354	-0.047861	-0.113807	-0.163964
C_{Yr}	0.000797	0.000903	0.020969	0.129108
$C_{l\beta}$	-0.060937	-0.061152	-0.079195	-0.068754
C_{lp}	-0.290752	-0.309150	-0.319870	-0.300537
C_{lr}	-0.022409	-0.020386	-0.008176	-0.004035
C_{ma}	-0.532747	-0.592106	-0.539750	-0.664614
C_{mq}	-1.573596	-1.867433	-1.699403	-2.055928
$C_{n\beta}$	0.002297	0.002213	0.005226	0.038278
C_{np}	-0.010008	-0.023346	-0.007755	0.023668
C_{nr}	-0.000151	-0.000937	-0.002450	-0.034222

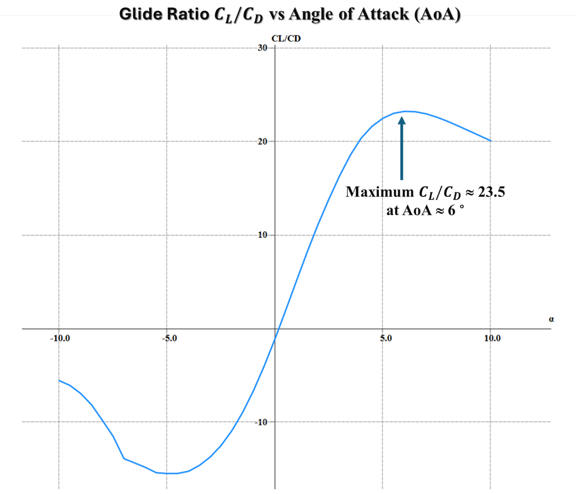
Table 4.8. Non-Dimensional Stability Derivatives in the Desired Order

Mode	Parameter	Design 0	Design 1	Design 2	Design 3
Phugoid	ω_n (Hz)	0.072	0.065	0.061	0.054
	Damped ω_d (Hz)	0.072	0.065	0.061	0.054
	Damping Ratio ζ	0.003	0.003	0.003	0.005
Short-Period	ω_n (Hz)	1.844	1.974	1.870	2.233
	Damped ω_d (Hz)	1.697	1.830	1.746	2.113
	Damping Ratio ζ	0.390	0.375	0.358	0.323
Roll Mode	Time to half (s)	0.200	0.180	0.213	0.305
	Time constant (s)	0.289	0.260	0.308	0.440
Dutch-Roll	ω_n (Hz)	0.218	0.243	0.405	0.443
	Damped ω_d (Hz)	0.216	0.240	0.405	0.442
	Damping Ratio ζ	-0.150	-0.156	0.020	0.050
Spiral Mode	Time to half (s)	122.507	92.652	40.083	13.933
	Time constant (s)	176.740	133.669	57.828	20.101

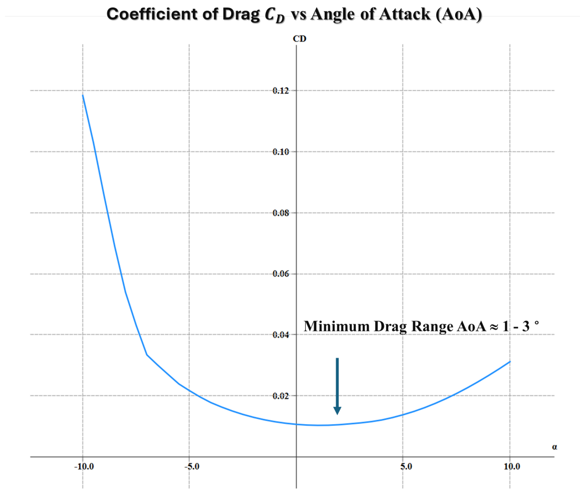
Table 4.9. Dynamic-Stability Parameters

The tables above demonstrate the progressive refinement of both static and dynamic stability characteristics across each design iteration. Improvements are especially evident in yaw and pitch damping, with Design 3 exhibiting significantly enhanced lateral-directional and longitudinal responses. Notably, the increased values of $C_{n\beta}$ and C_{mq} indicate a more stable and responsive configuration. To further validate final static margin and trim behaviour, the next section presents pitching moment (C_m) versus angle of attack (α) and lift coefficient (C_L), from which the trim point and overall static longitudinal stability will be confirmed.

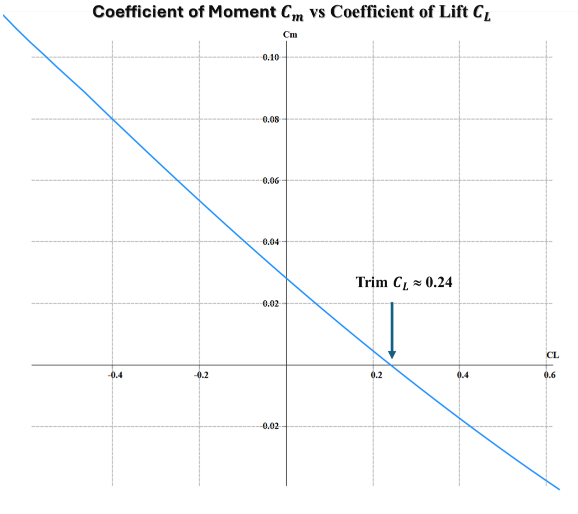
Finally, Figure 4.10 illustrates the static stability and performance analysis of Design 3. It achieves a maximum glide ratio (L/D) of approximately 23.5 at 6° angle of attack, indicating strong aerodynamic efficiency even with winglets. Minimum drag occurs between 1–3° AoA, with a trim angle at 3.9° for a centre of mass 0.5 m aft of the leading edge. A positive C_L at trim (0.34) confirms satisfactory lift generation in steady, trimmed flight. While the trim point lies slightly below the optimal L/D condition, small CoM changes or added twist can move it closer to peak efficiency. However, such changes must be cautiously tested to preserve the improved dynamic stability profile. The aerofoil configuration used is the sequence of Fauvel 14%, Horten 13%, and MH60 sections across the span. As shown in subsubsection 4.4.2, all tested root configurations were viable, reinforcing that aerodynamic performance is less sensitive to aerofoil arrangement than to dynamic stability characteristics.



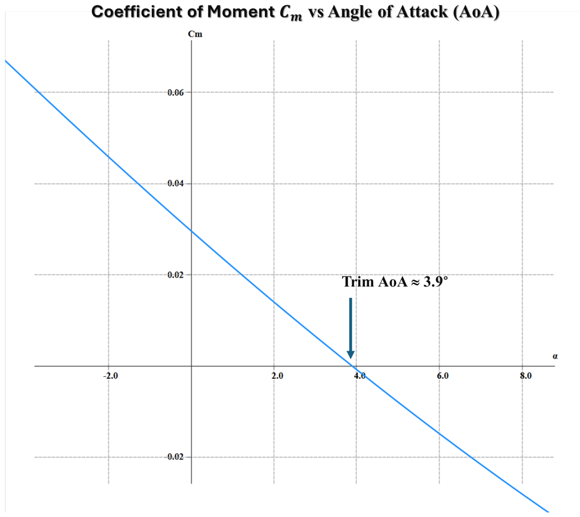
(a) Lift-to-Drag Ratio vs. Angle of Attack.



(b) Drag Variation C_D) vs. Angle of Attack.



(c) Lift Coefficient C_L vs. C_m



(d) Pitching Moment C_m vs Angle of Attack.

Fig. 4.10. Static Stability and Aerodynamic Performance of Design 3.

4.6 Propulsion System and Performance Evaluation

Following the constraints analysis in Section 4.2, an optimum T/W ratio of 0.155 was found at the maximum permissible aircraft weight of 25 kg. Two possible propulsion systems considered in this section are the XICOY X45 (45 N max thrust) and the JetsMunt Merlin M70XBL (70 N max thrust).

The X45 is the main contender, providing 45 N of thrust and yielding a T/W ratio of 0.183 at maximum take-off weight. This exceeds the 39 N minimum thrust required for level flight and satisfies the system's performance demands. Initially, the higher thrust XICOY X60 was intended for comparison, but due to limitations in efficiency data, the M70 was selected instead. Nevertheless, the M70 offers higher maximum thrust and potential for improved thrust-specific fuel consumption (TSFC) at lower throttle settings during cruise. Both engines are evaluated to determine their overall performance characteristics, with final selection based on their ability to support the mission objective of maximising range and endurance.

4.6.1 Thrust-Specific Fuel Consumption (TSFC) Comparison

4.11 and 4.12 reveal a marked difference in where each engine reaches peak efficiency. The X45 achieves its minimum TSFC at 35 N, while the M70 does so at 53 N, with both engines performing best at around 3/4 open throttle. Ultimately, the ideal engine selection will depend on the thrust required to cruise for best aerodynamic performance and will require further analysis.

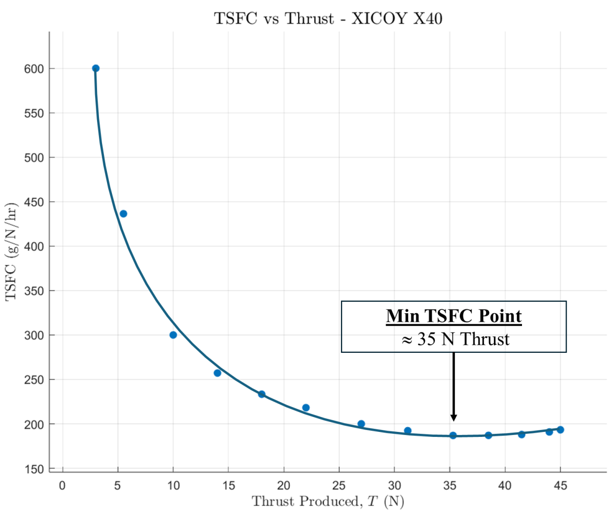


Fig. 4.11. TSFC at Different Thrust Points for the XICOY X45 Engine

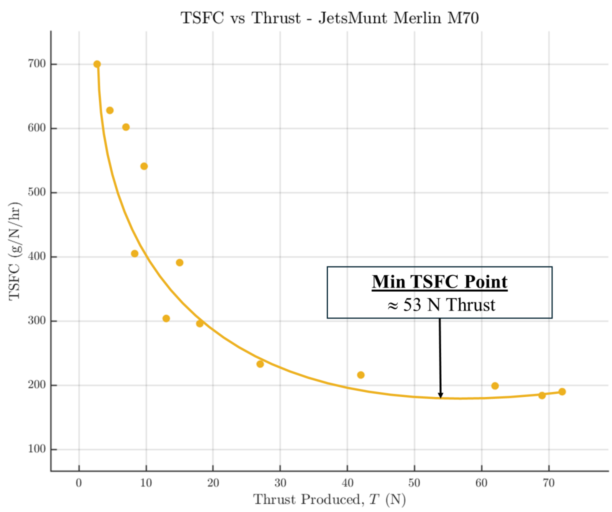


Fig. 4.12. TSFC at Different Thrust Points for the Merlin M70 Engine

4.6.2 Thrust and Velocity Considerations for Efficient Cruise

To determine the most suitable engine for a mission focused on range and endurance, it is necessary to identify the cruise velocity that optimises each. Figures 4.13 and 4.14 present the range and endurance characteristics of the final stable aircraft configuration, Design 3 (from Section 4.5), with the CoM positioned 0.5 m aft of the leading edge. It is clear from this analysis that the conditions for maximum range and maximum endurance do not co-incide. The maximum range occurs near 38 m/s, where the aircraft travels the furthest per unit of fuel consumed, while maximum endurance is attained at about 28 m/s, so favouring lower throttle and thrust for prolonged airtime.

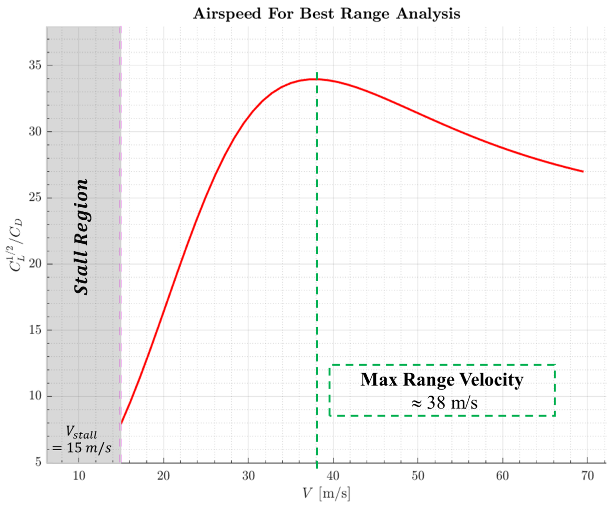


Fig. 4.13. Analysis of Cruise Airspeed to Maximise the Range Parameter $C_L^{1/2}/C_D$

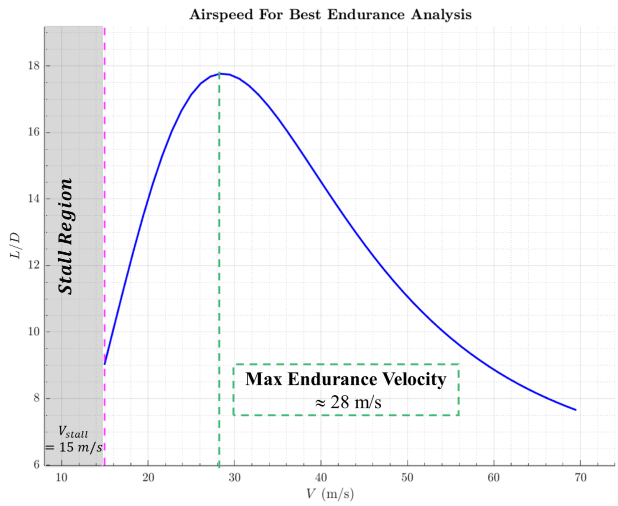


Fig. 4.14. Analysis of Cruise Airspeed to Maximise the Endurance Parameter C_L/C_D

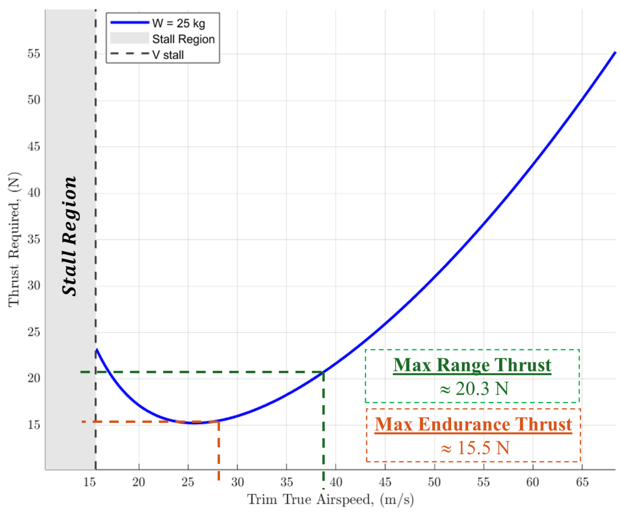


Fig. 4.15. Thrust Required vs. Trim True Airspeed at Maximum Weight.

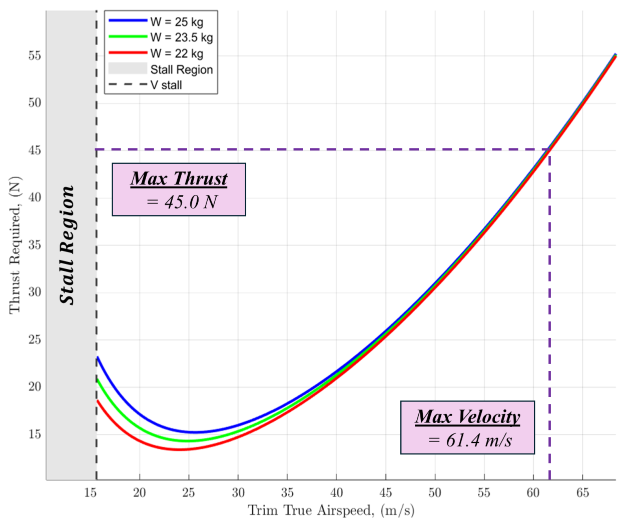


Fig. 4.16. Thrust Required vs. Trim True Airspeed at Different Aircraft Weights

Figure 4.15 shows the thrust required to achieve trim at cruising airspeeds, while Figure 4.16 presents the changes in this requirement with respect to the aircraft weight. At maximum weight, maintaining the endurance speed of 28 m/s requires about 15.5 N of thrust, while the maximum-range speed of 38 m/s requires around 20.3 N. These thrust levels make engine efficiency at these operating points of critical importance. As is evident in Section 4.6.1, the Merlin M70XBL is at a disadvantage as its TSFC at 15.5 N is roughly 315 g/N/hr compared to the X45's 250 g/N/hr, which makes the X45 appropriate for an endurance optimised mission. To achieve maximum range, the aircraft requires 20.3 N of thrust, which is supplied by the X45 at a TSFC of 216 g/N/hr whereas the M70 does so at 285 g/N/hr.

Lastly, Figure 4.16 displays the variation in thrust demand with respect to weight, also highlighting its maximum capable velocity. It is clear that even less thrust is required at lower weights and the X45 remains superior as it performs well across a range fuel loads, at both maximum endurance and range velocities. In addition, it achieves a top speed of 61.4 m/s, which is more than sufficient to meet the operational requirements of this mission.

4.6.3 Specific Air Range and Endurance

Specific Air Range (SAR) measures the distance an aircraft can fly per unit of fuel burned (m/kg) , while Specific Endurance (SE) quantifies the flight time achieved per unit of fuel (min/kg). The following section analyses both SAR and SE for 'Design 3' across a range of cruise airspeeds and aircraft weights to identify the speeds that maximise distance travelled and time in the air for our endurance-focused mission profile.

The XICOY X45 engine consumes kerosene + 4% oil, commonly known as Jet A-1 fuel. For this design, the aircraft carries a 4 L fuel tank, equating to approximately 3.2 kg of fuel. Additionally, an important factor which strongly influences both SAR and SE is the input TSFC, which has been selected as 233 g/N/hr by averaging both the endurance and range TSFC values from Section 4.6.2, to ensure a fair comparison. Figures 4.17 and 4.18 illustrate specific air range and endurance over a range of trim cruise airspeeds at three different masses, 25 kg (max take-off), 23.5 kg, and 22 kg, to reflect fuel burn. Although the MATLAB tool is able to account for altitude changes, this mission's 120 m service ceiling makes air-density variations negligible, so all results assume sea-level conditions. Assuming the aircraft consumes its entire 3.2 kg fuel load, Figure 4.17 shows that flying at 33.7 m/s maximises range, delivering about 30,000 meters per kilogram of fuel burnt, equivalent to 96 km on a full tank.

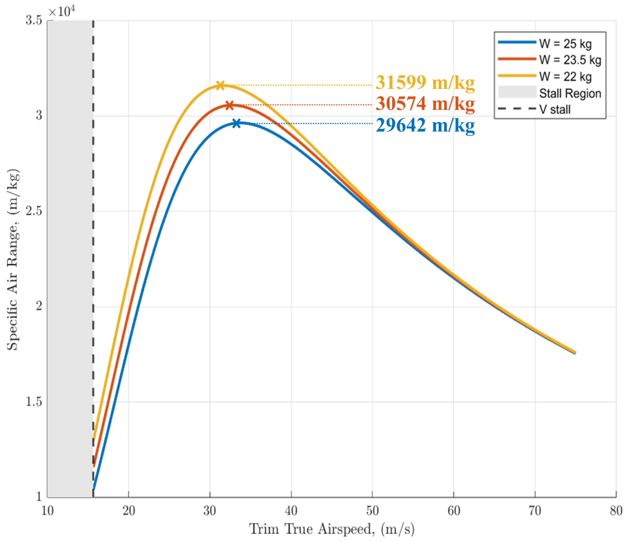


Fig. 4.17. Specific Air Range (m/kg) vs. Trim Airspeed (m/s)

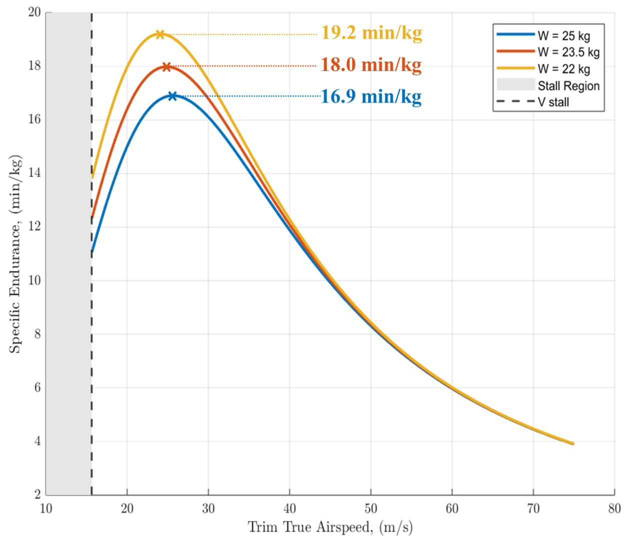


Fig. 4.18. Specific Endurance (min/kg) vs. True Airspeed (m/s)

Likewise, Figure 4.18 shows that flying at 25.6 m/s maximises endurance, yielding roughly 17 minutes per kg, which corresponds to 54.4 minutes of airtime when all 3.2 kg of fuel is used. It is also important to note that perfect efficiency cannot exist. Achieving both maximum endurance or range and minimum TSFC simultaneously would require an idealised engine and flight condition tailored exactly to each mission point. In reality, trade-offs are inevitable, and performance must be optimised around realistic conditions rather than ideal extremes.

TSFC Sensitivity Analysis

This sensitivity analysis assumes the aircraft is at its maximum take-off mass of 25 kg with a 3.2 kg fuel load. Table 4.10 displays how SAR and SE respond to ±20% variations in TSFC, applied in steps of 5% with respect to the nominal value of 233 g/N/hr. At –20% TSFC, SAR increases from 29,642 to 37,051 m/kg corresponding to an additional 25% increase in range (23.7 km extra range with maximum fuel). Additionally, SE increased from 16.9 to 21.1 min/kg extending total airtime by 24.8% (13.4 minutes extra with maximum fuel). Conversely, at +20% TSFC, SAR falls by 16.6% to 24,701 m/kg (≈16 km less range overall) and SE decreases by 16.5% to 14.1 min/kg (≈9 minutes less airtime overall). As further illustrated in Figure 4.19, the plot shows a nearly linear relationship between the percentage change in TSFC and the corresponding variations in SAR and SE, emphasising a predictable impact of TSFC variation.

Drag Coefficient Uncertainty and Sensitivity Analysis

FLOW5 and XFLR5 both use Vortex Lattice Method (VLM) to conduct aerodynamic anal-

ΔTSFC	TSFC (g/N/hr)	SAR (m/kg)	SE (min/kg)
–20 %	186.4	37 051	21.1
–15 %	198.1	34 723	19.8
–10 %	209.7	32 935	18.8
–5 %	221.4	31 194	17.8
0 %	**233.0**	**29 642**	**16.9**
+5 %	244.7	28 224	16.1
+10 %	256.3	26 947	15.4
+15 %	268.0	25 770	14.7
+20 %	279.6	24 701	14.1

Table 4.10. Impact of TSFC variation on SAR and SE.

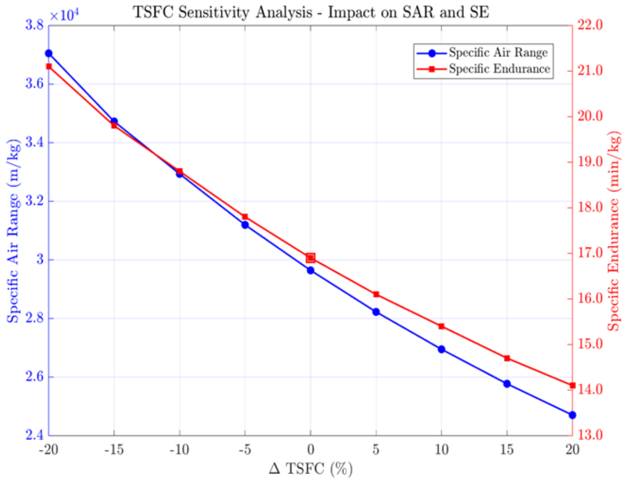


Fig. 4.19. Relationship Between TSFC Changes and SAR/SE Performance.

yses which neglect viscous effects, leading to an underestimated drag coefficient by roughly 20% (XFLR5, 2025). This uncertainty is the driving factor for this sensitivity study of C_D. Table 4.11 shows how SAR and SE respond to incremental increases in C_D from the nominal value of 0.014. At +20% C_D, SAR falls from 29,642 to 25,854 m/kg, corresponding to a 12.8% reduction in range (12.1 km less range with maximum fuel load), and SE decreases from 16.9 to 15.4 min/kg, an 8.8% reduction and roughly 5 minutes less total flight time.

ΔC_D (%)	C_D	SAR (m/kg)	SE (min/kg)
0 %	**0.0140**	**29 642**	**16.9**
+5 %	0.0147	28 577	16.5
+10 %	0.0154	27 597	16.1
+15 %	0.0161	26 692	15.8
+20 %	0.0168	25 854	15.4

Table 4.11. Impact of C_D variation on SAR and SE.

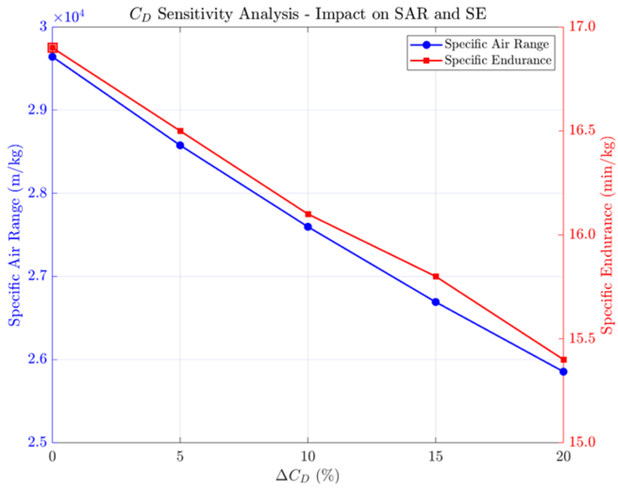


Fig. 4.20. Relationship Between C_D Changes and SAR/SE Performance.

Compared to the TSFC sensitivity, the drag uncertainty produces smaller performance variations. In fact, a +10% increase in TSFC reduces SE to 15.4 min/kg, equivalent to the SE at +20% C_D, showing that the drag coefficient's effect on endurance is roughly half as sensitive as TSFC. Figure 4.20 illustrates a similarly linear relationship compared with the TSFC analysis, with a less steep gradient, further signifying its predictable impact.

4.7 Operating Range Analysis of Final Design

This section evaluates the operational limits of 'Design 3', analysing its longitudinal stability, center-of-mass range, and key flight performance parameters. Figure 4.21 shows the required elevon deflection angles to maintain trim at different airspeeds and center-of-mass (CoM) locations. The slope of this relationship indicates the stability characteristics. A positive slope (nose-up trim) means the aircraft is statically stable; a zero slope marks the location of the neutral point; while a negative slope would show static instability, meaning the aircraft would pitch up uncontrollably unless the pilot applies opposite elevator input, a sign of insufficient static margin.

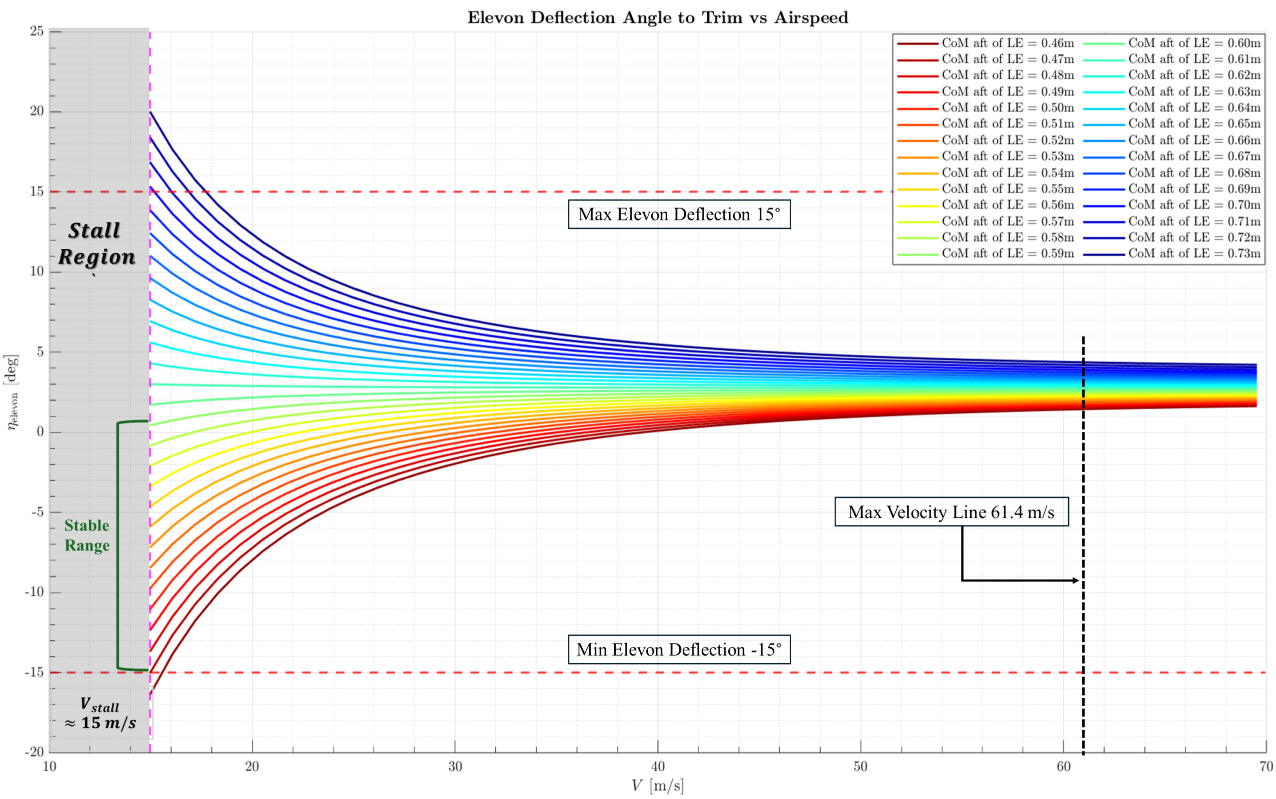


Fig. 4.21. Elevon Deflection Angle to Trim at Various CoM Locations vs. Airspeed.

Based on the elevon deflection analysis, the recommended center-of-mass (CoM) range is 0.47-0.59 m aft of the leading edge. Values below 0.47 m fall beyond elevon deflection lower limit (-15°), while positions beyond 0.59 m approach the instability boundary.

The next steps in the operational analysis involve utilising the design parameters of 'Design 3' as outlined in Table 4.12, which specifies the aircraft's weight characteristics, aerodynamic properties, and performance targets. These parameters were used as inputs in the MATLAB-based performance tool to calculate the values and ranges presented in Table 4.13.

Design 3 Parameters	Value	Unit
Max Take-off Weight	245.25	N
Max Available Thrust	45	N
Fuel Capacity	4	Litres
Max Empty Weight	213.86	N
Total Wing Area	1.560	m^2
Projected Wing Area	1.364	m^2
Max L/D AoA	6	deg
Max L/D	23.3	–
$C_{\mathrm{L,max}}$	1.2	–
$C_{\mathrm{L,TO}}$	0.5	–
$C_{\mathrm{D,0}}$	0.014	–
$C_{\mathrm{D,TO}}$	0.023	–

Table 4.12. Specifications and Aerodynamic Properties of 'Design 3'.

Parameter	Value	Baseline Algorithm
CoM Range From LE		(Flight Dynamics, 2025)
Forward Limit	0.47 m	
Aft Limit	0.59 m	
Ground Roll		(Gudmundsson, 2022)
Take-off Velocity (V Rotate)	17.2 m/s	
Take-off Distance	157.7 m	
Landing Distance	246.3 m	
Braking Distance	220.5 m	
Touchdown Speed	17.2 m/s	
Flight Envelope		(Gudmundsson, 2022)
Stall Speed	15.6 m/s	
Maximum Operating Speed	61.4 m/s	
Service Ceiling	120 m	
Optimal Performance Speeds		(APD, 2024)
Max Range	38 m/s	
Max Endurance	28 m/s	

Table 4.13. Operational and Performance Metrics Overview.

Ground performance includes a 157.7 m take-off distance at maximum weight and a 246.3 m landing distance, assuming 3 out of 4 litres of fuel have been consumed. The flight envelope shows a stall speed of 15 m/s and a maximum operating velocity of 61.11 m/s, with optimal performance at 38 m/s for maximum range and 28 m/s for endurance. These results provide an overview to help the operator understand the aircraft's operating ranges and performance.

All MATLAB codes used in this section and throughout the results are provided in Appendix C to ensure full reproducibility and facilitate further research.

5 Discussion

5.1 Project in the Broader Engineering Context

The methodology used for this project is in line with the demand for computational cost effective and easy-to-use tools to aid in the design of UAS in regulatory environments such as the UK Civil Aviation Authority (CAA) Open Category. By not using computationally intensive computational fluid dynamics (CFD) or high-end simulations that demand supercomputer capacity, the tool is a realistic solution that is primarily created towards small-scale UAS developers, university research teams, and hobbyists. This makes the performance analysis democratic in a way that allows stakeholders with minimum resources to design a stable aircraft for substantial range and endurance parameters. The approach follows the use of standard, widely available computing tools by relying on reduced-order aerodynamic models (such as XFLR5 and MATLAB) and simplified environmental and drag estimates. This makes the tool a practical middle ground between basic spreadsheet methods and expensive, time-consuming computational fluid dynamics. The code is also easily transferable to other programming languages, such as Python, making it more accessible to a wider range of users. Overall, the method addresses an important engineering need for fast and affordable UAS prototyping.

5.2 Project in the Wider Literature

The tailless flying wing configuration is still relatively under-explored within the small UAS context, especially for jet-powered aircraft under strict size limitations. Existing literature tends to favour either conventional tailed configurations or high-fidelity simulations unsuitable for quick design iteration. This dissertation addresses this gap by illustrating a low-order modelling methodology that employs XFLR5's Vortex Lattice Method (VLM) and three-dimensional panel approaches to accurately predict aerodynamic performance characteristics. Notably, XFLR5 also enables stability and dynamic response analysis, providing both low-cost modelling and essential flight-worthiness assessments.

Central to this dissertation is a MATLAB-based performance analysis tool that integrates reduced-order aerodynamic data from XFLR5's Vortex Lattice and 3D panel methods with analytical drag and propulsion models to automate the initial sizing and performance evaluation of tailless UAS. Beginning with user-defined requirements such as cruise velocity, the tool computes preliminary wing area and required thrust-to-weight ratio. It then applies the Breguet equations to determine optimal cruise speeds for maximum range and endurance. Furthermore, utilising empirical relations to predict take-off and landing distances at varying weights. Static stability margins (longitudinal, lateral, directional) are calculated from XFLR5 stability derivatives, ensuring compliance with CAA Open Category criteria.

By automating these steps, the tool enables rapid design iteration, faster than high-fidelity CFD, while retaining sufficient accuracy for early-stage concept selection. This hybrid methodology aligns with recent UAS research advocating model-based prototyping that balances computational efficiency and fidelity. Furthermore, the modular MATLAB code is easily transferable to Python or other languages, lowering barriers for future users. In filling the gap between simplistic spreadsheet approaches and resource-intensive simulations, this dissertation provides a practical, accessible environment for maximising range, endurance, stability and ground performance of jet-powered tailless UAS under realistic regulatory constraints.

5.3 Limitations of the Methods and Impacts

XFLR's 3D panel and Vortex Lattice Method (VLM) analyses neglect viscous effects, leading to underestimated total drag and over predicted performance. While XFLR5 does compensate to some extent by interpolating 2D viscous drag values, there is no way this can account for complicated 3D flow separation or boundary layer interactions, particularly at high angles of attack or on very swept tailless wings. Consequently, total drag is typically underestimated by around 20%, leading to glide-ratio estimates and performance metrics that are mildly optimistic.

Equally, the planar-quadrilateral panel formulation constrains the geometric fidelity with which curved or blended flying-wing surfaces can be represented. Fine details of surface curvature and twist must be approximated by flat facets, introducing small errors into the computed pressure distributions and stability derivatives. While these geometric simplifications affect absolute coefficient values, they do so in a consistent manner across all design variants, allowing the tool to identify relative performance differences.

Finally, XFLR5's numerical algorithms exhibit sensitivity to Reynolds number outside the mid-range regime. Predictions may exhibit inaccuracies below Reynolds numbers of approximately $\times 10^4$, where viscous forces dominate, and convergence issues can arise above about 1 $\times 10^6$, with high-Re solutions occasionally becoming unstable or diverging. In practice, however, small UAS operating in the UK CAA Open Category generally fall within the tool's reliable Reynolds-number window, and XFLR5 remains accurate up to its convergence limits.

Although the case study presented in this report may yield somewhat optimistic performance parameters for range, endurance, and take-off/landing distances, the underlying performance tool remains robust—particularly when accurate aircraft specifications are available. In circumstances where such data are calculated using software such as XFLR5, users should account for the tool's optimistic tendency in their analysis. The performance tool remains valid for its intended role of revealing trends and supporting reliable relative comparisons during the preliminary design phase. For low-weight, medium-speed platforms typical of UK CAA Open Category operations, its fidelity is more than acceptable. However, the application of the methodology employed in this dissertation to aircraft designs that experience either very high or very low Reynolds number extremes, such as those encountered in significantly larger or higher speed aircraft, would necessitate the adoption of more advanced methods, such as high-fidelity CFD.

5.4 Discussion of Results and Implications for Aims/Objectives

The primary aim of this project was to develop a toolchain capable of optimising range, endurance and stability. This tool was to then to be applied to a case study design of a tailless, jet-powered UAS while remaining compliant with the UK CAA Open Category regulations. This aim was successfully achieved through a structured and iterative design process, guided by the following key objectives:

Objectives and Key Results

Objective 1: Design Requirement Specification

The initial constraint analysis determined a feasible design space with a thrust-to-weight ratio $\geq$ 0.159 and wing loading between 100–200 $\mathrm{N/m^2}$, corresponding to a wing area range of 1.23–2.45 $\mathrm{m^2}$ and at least 39 N of maximum installed thrust. This ensures the rest of the design is within the policy and regulations MTOW $\leq$ 25 kg and ceiling $\leq$ 120 m.

Objective 2: Aerodynamic Configuration Optimisation

Six reflexed aerofoils were compared in XFLR5/FLOW5, scoring for number of require parameters for the mission purpose including lift-to-drag, stall behaviour and trim moment. The MH60 profile scored highest across different conditions. It was then paired with Fauvel 14% and Eppler 339 and Horten 13% to form the flying-wing planform.

Objective 3: Geometry and Stability Optimisation

Iterative geometry modifications, addition of a cambered Fauvel 14% winglet, 5° anhedral, washout and CoM placement at 0.5 m aft of the leading edge, produced "Design 3," which balanced high L/D (23.5 at 6° AoA) against static and dynamic stability requirements. Stability derivatives and dynamic modes all met then minimum requirements and beyond: C_{m_α} from -0.540 to -0.665, $C_{n_\beta} = +0.0383$, Dutch-roll damping $\zeta = 0.05$, and adequate time-to-half for phugoid, short-period, roll and spiral modes. This confirms the tailless configuration achieves both static and dynamic stability without a tailplane.

Objective 4: Propulsion System Choice

The XICOY X45 micro-turbine was selected for its low TSFC at both endurance (216 g/N/hr at 28 m/s) and range (250 g/N/hr at 38 m/s) cruise points, matching the thrust requirements (15.5 N and 20.3 N respectively).

Objective 5: Integrated Performance Modelling

Using the MATLAB toolchain, the final design achieved a Specific Endurance of 16.9 min/kg (≈54 min total with 4 L fuel) and a Specific Air Range of 2.96 km/kg (≈95 km total), validating the tool's ability to predict mission performance metrics.

Objective 6: Final Design Operating Range

"Design 3" meets all primary aims: compliance with MTOW and service ceiling; maximum range/endurance; and necessary stability margins. The CoM adequate stable range (0.47 - 0.59 m aft of leading edge). Ground-roll performance (take-off 158 m, landing 246 m) and envelope (stall 15.6 m/s, V_{Max} 61.4 m/s) further confirm operational viability.

Sensitivity and Future Refinements

Although the design meets its desired objectives, the drag coefficient may be underestimated by up to 20%, and TSFC variations of ±20% were shown to affect the specific range and endurance by ≃16%. Since the aim was not only to optimise range, endurance, and stability but also to accurately predict them, these sensitivities reveal key limitations in predicted performance. However, the results remain valuable and illustrates the importance of conducting sensitivity and uncertainty analyses. Future work should focus on validating and refining these predictions to better understand their practical significance.

6 Conclusion

The goal of this dissertation has been to conceptualise and design a tailless, UAS that complies with the UK CAA's Open Category guidelines. The methodology implemented to meet the project aims and objectives involves performing an initial sizing analysis to determine feasible wing areas and thrust requirements. This analysis forms the subsequent design processes. A selected series of reflexed cambered aerofoil profiles are then examined on a flying-wing planform, assessing their aerodynamic efficiency and stability to ensure the theoretical design satisfies the mission requirements and safety regulations. This conceptual design is further used to aid the developed tool by computing performance parameters such as take-off, landing and optimal cruise endurance and range metrics.

The propulsion system selected is the XICOY X45 micro-turbine, due to its low thrust-specific fuel consumption (TSFC) at trim cruise airspeeds that maximise endurance and range. The aircraft operates efficiently at cruise velocities optimised for both endurance and range, meeting thrust requirements while maintaining fuel efficiency. Furthermore, computation of specific air range (SAR) and specific endurance (SE) are evaluated at maximum weight. The resulting SE is 16.9 min/kg, equating to 54 minutes of flight time at maximum fuel capacity of and a maximum specific air range of 2.96 km/kg corresponding to roughly 95 km on a full tank.

The final design configuration, denoted as "Design 3," achieved an estimated maximum lift-to-drag ratio of 23.5 angle of attack, demonstrating strong aerodynamic efficiency for an aircraft of this scale and layout. The final refinement from 'Design 2' to the 'Design 3' involved the addition of a cambered Fauvel 14% winglet as well as anhedral, significantly enhancing lateral and longitudinal stability as well as dynamic modes. Improvements were observed in directional stability, static stability and pitch rate damping, smoothing the short-period oscillations. By this stage, all principal modes, including phugoid, short-period, roll, Dutch-roll and spiral met sufficient time to half and damping, confirming that the final wing geometry delivers the required static and dynamic stability for a tailless UAS without a conventional tailplane. To develop the stability study further, recommended research would involve incorporating 'non-linear time-domain' simulations of the aircraft to more accurately illustrate control authority and focus on how the aircraft reacts to sudden wind changes during flight.

Based on the finding of this research project, the proposed design successfully meets the performance and stability objectives for a tailless, jet powered UAS. However, analysing the sensitivity and uncertainty analyses reveals important areas for improvement. The examined area highlights a substantial impact on the SAR and SE, when the drag and TSFC are altered. Given that the drag coefficient used throughout this study may be underestimated by up to 20%, further clarification of aerodynamic analysis methods is recommended. High-fidelity CFD simulations or wind tunnel testing could aid this process and reduce this uncertainty, providing a more reliable prediction of real-world performance. Additionally, the XICOY X45's TSFC sensitivity analysis revealed that propulsion efficiency has a substantial influence on mission performance metrics, as even a 20% increase can influence the overall range and endurance roughly by 16%. To build on this insight, future work recommendation includes real-world testing of the X45 alongside other micro-jet turbines. This will aid capturing accurate fuel consumption characteristics across various mission sections potentially revealing a more efficient engine candidate for the mission.

Alongside these performance results, the development of a flexible MATLAB based performance analysis contributes to the early-stage design of unconventional UAS literature. The toolchain bridges the gap between traditional, low-fidelity spreadsheet methods and resource-intensive CFD analyses. This offers a quick and adaptable solution that allows small research teams, independent developers, and hobbyists to perform studies and designs with minimal computational demands. Furthermore, the flexible structure of the tool enables adaptation to broader aerospace applications beyond the original scope. By modifying the propulsion and flight dynamic models, as well as flight conditions, the framework can be extended to support electric systems, blended wing body configurations, or even conventional aircraft layouts. The basic principles will always remain the same, making it a practical resource for both academic research and practical design work. This enables studies that go beyond the immediate scope of this project. Moreover, even though the prescribed mission profile is strictly limited by the CAA's regulations of the Open Category, requiring Visual Line of Sight (VLOS) operations, adjusting to more real world applicable operational modes within the Specific Category can be done by varying the design framework. Such operational modes can include beyond-Visual Line of Sight (BVLOS) flights or even larger aircraft. The flexibility of this design allows it to be applied to environments beyond those originally discussed in this study. Such missions could include search-and-rescue operations, urban mapping and environmental studies in remote areas demanding longer operational durations and coverage.

7 Management

7.1 Project Plan

Effective time management was essential to the success of the project, particularly given the need to balance alongside other academic commitments. This is reflected by the series of Gantt charts included in Appendix A, which illustrate the evolution of the project timeline over the two semesters.

The initial Gantt chart produced at the start of Semester 1 (Figure A.1) was overly ambitious and lacked realistic alignments and tasks required to conduct the research project. It also did not sufficiently account for the workload from other modules, such as group projects and coursework submission. As a result, tasks began to fall behind schedule early on. This necessitated a revised Gantt chart to be developed later in Semester 1 (A.2), which detailed a clearer more achievable plan, factoring in commitments from other units.

In Semester 2, the project plan was further developed (Figure A.3) as the project objectives became more clear, detailing more specific tasks to do, continuing to ensure enough time is allocated to other unit commitments. A final, detailed Gantt chart produced in April (Figure A.4) provided specific tasks remaining ensuring it has enough detail to complete the remaining tasks over the Easter break.

7.2 Project Plan Reflection

The management strategy was successful to an extent; it clearly highlighted out all required tasks required for the success of the project. However, a great deal of the work still had to be pushed into the Easter break due to earlier schedule delays. This was mainly due to certain tasks requiring more time to complete than initially expected.

Implications on Future Projects

To better accommodate and plan for inevitable delays, future projects will incorporate the following strategies:

- **Effective Time Allocation:** I will assign 'buffer time' to each task based on estimated completion time, allocating even more time and pre-planning for simulation and data processing tasks.
- **Timeline Updates:** Update the Gantt chart more frequently (every two weeks) to reflect actual progress. Further adding extra tasks which may come up throughout the process.

7.3 Risk Assessment

As this project did not involve physical experimentation, the primary risks identified were related to mental well-being, data security and potential remote work hazards.

A completed General Risk Assessment Form from The University of Manchester is included in Appendix B, documenting their potential hazards and mitigation strategies.

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Appendices

A Project Planning

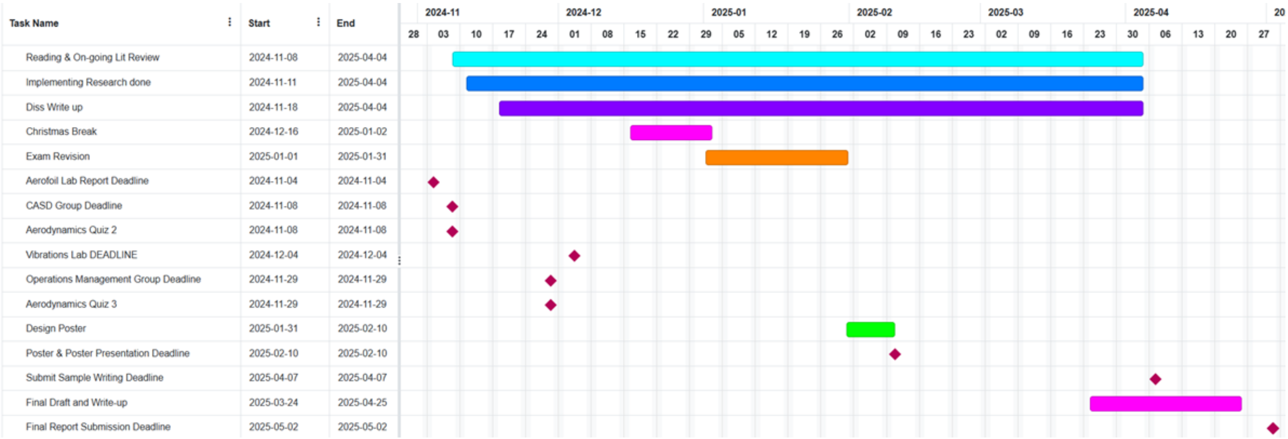

Fig. A.1. Semester 1 Initial Gantt Chart.

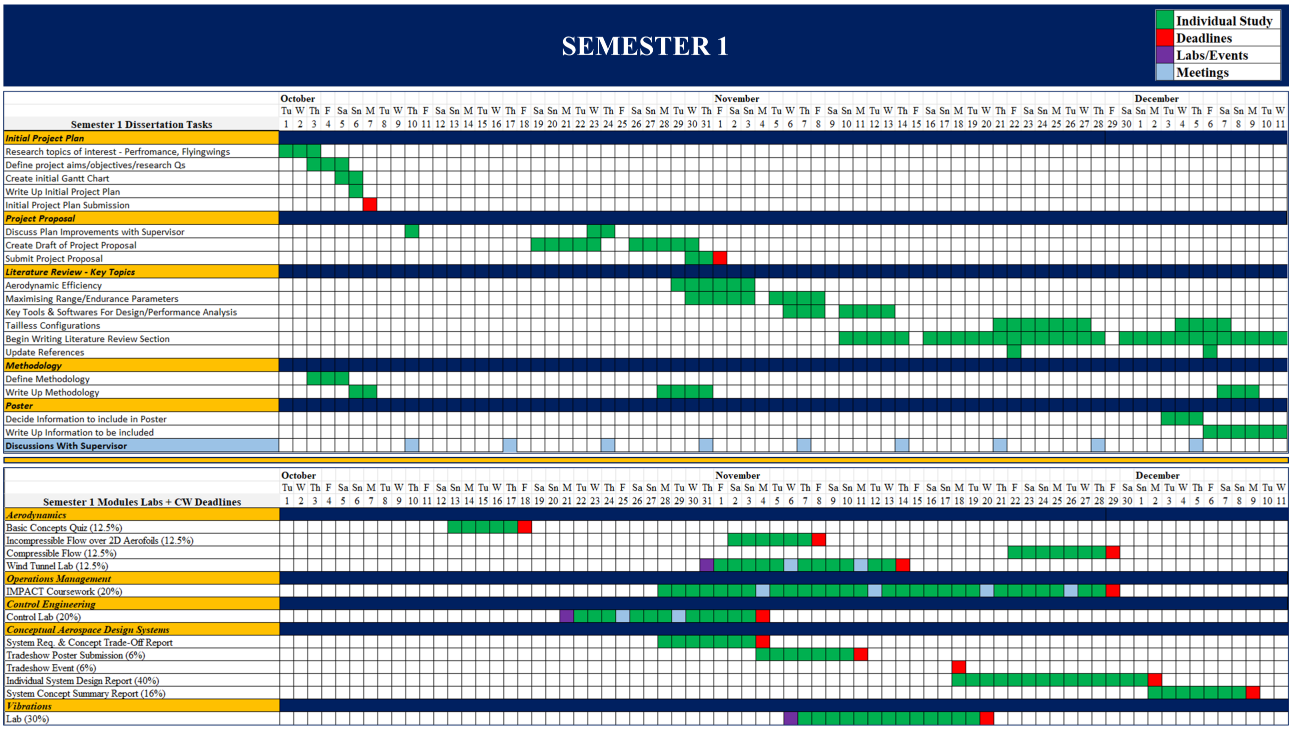

Fig. A.2. Semester 1 Updated Gantt Chart.

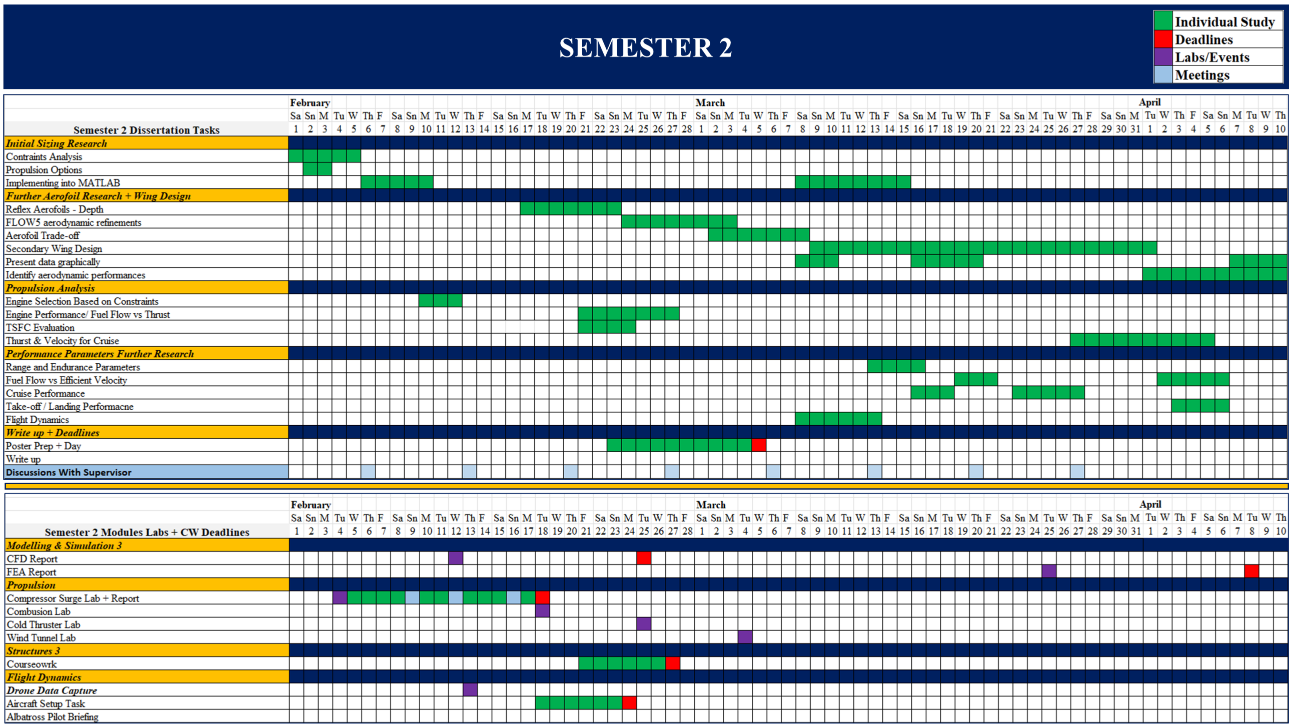

Fig. A.3. Semester 2 Gantt Chart.

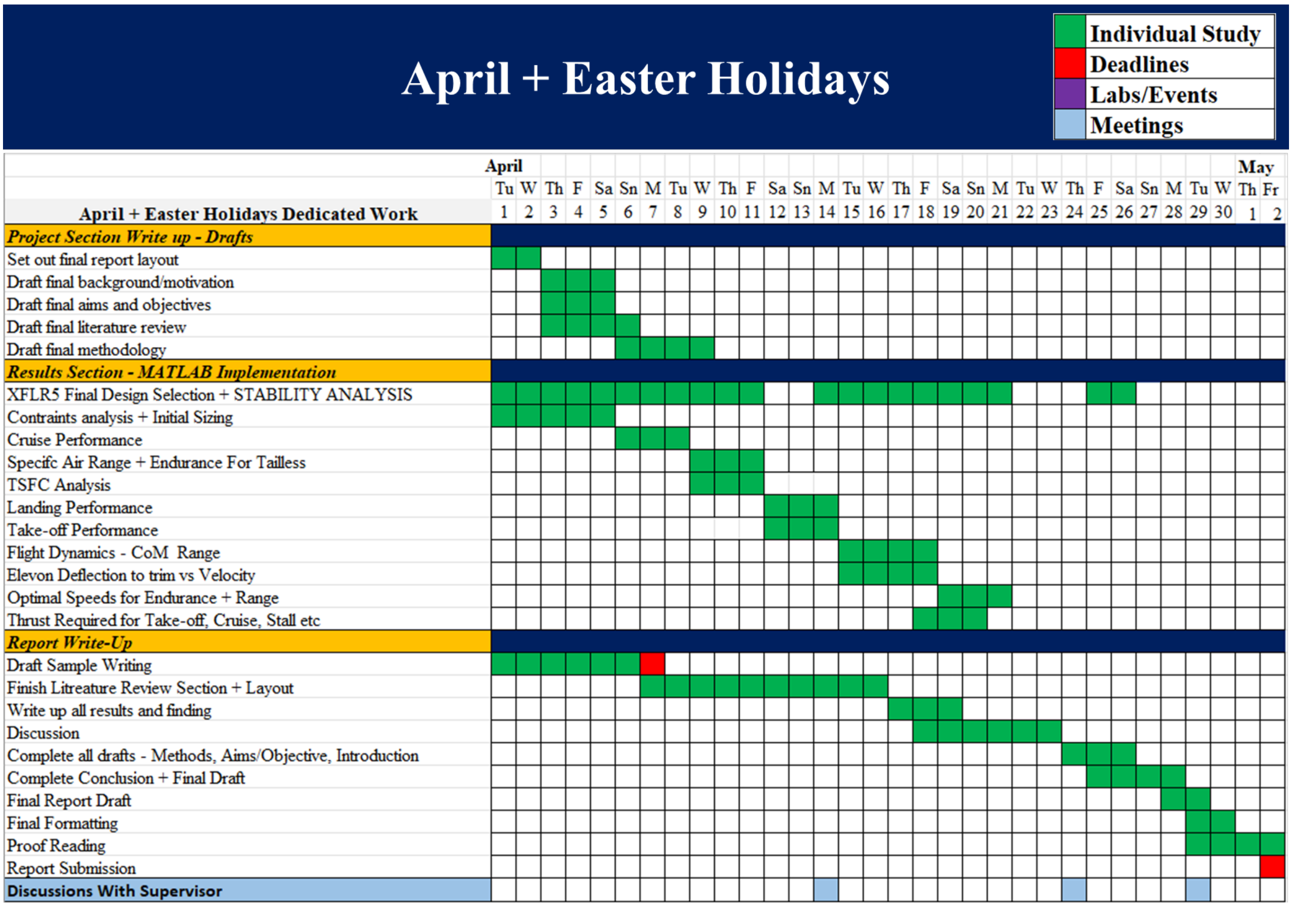

Fig. A.4. April Gantt Chart.

B Relevant Materials

Class of Aircraft	Range of C_{Dmin}	
	Lower	Upper
World War I Era Aircraft 1914-1918	0.0317	0.0771
Interwar Era Aircraft 1918-1939	0.0182	0.0585
Multi-Engine WW-II Bombers (Piston)	0.0239	0.0406
Single and Multi-Engine WW-II Fighters (Piston)	0.0157	0.0314
Multi-Engine Commercial Transport Aircraft (Piston)	0.0191	0.0258
Kitplanes (Homebuilt) and LSA	0.0119	0.0447
Single Engine GA Aircraft (Piston and Turboprop)	0.0174	0.0680
GA ag-aircraft, single engine, propeller, clean	0.0550	0.0600
GA ag-aircraft, single engine, propeller, spray-system	0.0700	0.0800
Twin Engine GA Aircraft (Piston and Turboprop)	0.0230	0.0369
Flying Boats	0.0233	0.0899
Selected Jet Fighter/Trainer Aircraft	0.0083	0.0240
Selected Jet Bomber and Attack Aircraft	0.0068	0.0160
Commercial Jetliners and Business Jets	0.0160	0.0219
Various Subsonic Military Aircraft (Props and Jets)	0.0145	0.0250
High performance sailplane	0.0060	0.0100
Tailless aircraft, propeller	0.0150	0.0200
Tailless aircraft, jet	0.0080	0.0140
Low altitude subsonic cruise missile (high W/S)	0.0300	0.0400

Fig. B.1. Typical Range of Subsonic Minimum Drag Coefficients, C_{Dmin} (Gudmundsson, 2022).

Kitplanes (Homebuilt) and LSA									
Make	Model	AR	S, ft²	S_{wet}, ft²	f, ft²	C_f	C_{Dmin}	LD_{max}	Reference
Bede	BD-5B	9.75	47.4	170	0.80	0.0048	0.0169	-	Stinton
Cozy	Mark IV	8.94	88.3	-	1.78	-	0.0201	16.5	Author
Falco	F.8L (160 & 180 BHP)	6.41	107.5	-	2.28	-	0.0212	14.2	Author
Flight Design	CTSW	7.29	107.4	-	3.64	-	0.0338	11.9	Author
Glassair	III (Standard Wing)	6.68	81.3	-	1.86	-	0.0229	14.0	Author
Icon	A5 (amphib)	9.03	128	-	5.72	-	0.0447	11.1	Author
Osprey	2 (amphib)	5.20	130	-	5.31	-	0.0408	9.5	Author
	GP-4	5.54	104	-	1.23	-	0.0119	18.0	Author
Progressive Aerodyne	Searey (amphib)	6.06	157	-	6.93	-	0.0441	9.7	Author
Pipistrel	Virus 912 LSA	14.10	118	-	1.50	-	0.0127	23.9	Author
Rutan	LongEz	8.31	82.0	-	1.88	-	0.0230	16.2	Author
	VariEze (wheelpants)	9.28	53.6	-	1.11	-	0.0207	16.5	Author
	VariEze (no whlpnts)	9.28	53.6	-	1.21	-	0.0226	15.8	Author
	VariEze	9.20	53.6	260	1.25	0.0047	0.0233	-	Stinton

Fig. B.2. Typical Characteristics of Selected Homebuilt Aircraft (Gudmundsson, 2022).

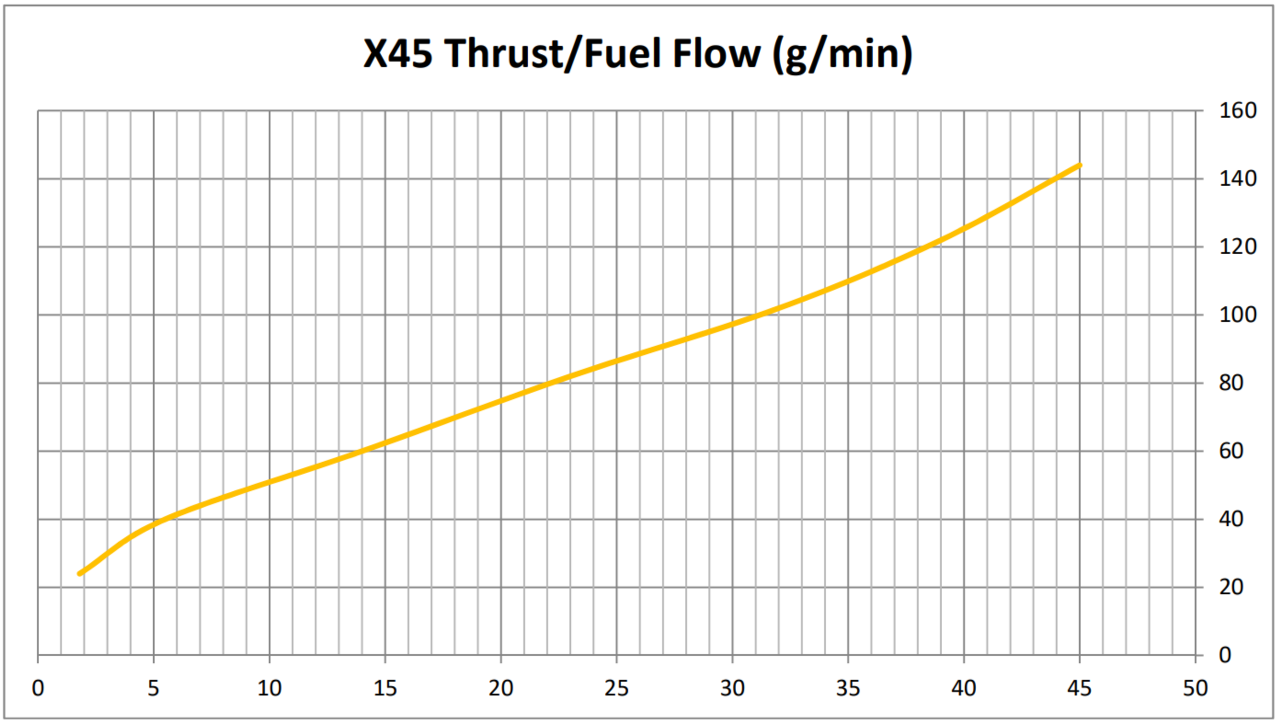

Fig. B.3. XICOY X45 Engine: Fuel Flow vs Thrust (XICOY, 2024).

Figure B.3 aided a more accurate TSFC plot conducted in Section 4.6.1, by providing manufacturer sourced thrust and fuel flow data, enabling a more validated propulsion model used in the performance analysis.

General Risk Assessment Form

Date: (1)	Assessed by: (2)	Checked / Validated* by: (3)	Location: (4)	Assessment ref no: (5)	Review date: (6)
29/04/2025	**Pedram Habib Zadeh**	**Khristopher Kabbabe**	**Remote**	**-**	**-**
Task / premises: (7) Dissertation Work					

Activity (8)	Hazard (9)	Who might be harmed and how (10)	Existing measures in place to control risk (11)	Risk rating (12)	Result (13)
Connecting laptop to mains power	Electrocution from exposed wire	Me – electric shock injury	▪ Pre-use inspection of cables ▪ Keep area dry and clear of any liquids	High	A
Prolonged work sessions	Mental health	Me – stress, burnout, anxiety	▪ Take regular breaks ▪ Task planning and management	Medium	A
Data loss/theft	Loss of results and work progress	Me – loss of important data	▪ Daily backups to a cloud ▪ Secure passwords ▪ Frequent backups on a memory drive	High	A
Sitting at workstation	Strain and discomfort	Me – back/neck injury	▪ 5–10-minute breaks every hour ▪ Comfortable seating chair with back rest	Medium	A
Working in the computer clusters	Fire hazards	Me – burns, smoke inhalation	▪ Fire escapes and route known ▪ Smoke and heat detectors ▪ Fire extinguisher nearby	High	A

Result : T = trivial, A = adequately controlled, N = not adequately controlled, action required, U = unknown risk

C MATLAB Code

The following link directs to a public GitHub repository containing the full MATLAB-based performance analysis tool developed for this project. It includes all necessary performance analysis scripts along with a README file for guidance:

Performance Tool Repository

If the link does not work, please visit https://github.com/pzxdeh/Performance-Tool.